| T.B.C. : CDS-MATH-ALL-24-A<br>Serial No. **940281** | **TEST BOOKLET**<br>**[ A ]**<br>**COMPLETE REPOSITORY** | **Comprehensive Archive**<br>**Marks Scheme: +1.0 / -0.333** |
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# UNION PUBLIC SERVICE COMMISSION

## COMBINED DEFENCE SERVICES (CDS) EXAMINATION

## ELEMENTARY MATHEMATICS — COMPLETE CHAPTER-WISE PYQ ARCHIVE

**1.** This question paper repository contains **ALL 707 Questions** extracted directly from official UPSC Combined Defence Services (CDS) examination archives.

**2.** Questions are organized into all official syllabus chapters with original item prompts and 4 alternative options.

**3. SCORING FORMULA:** +1.0 marks per correct response | -0.333 marks penalty for incorrect responses.

**4. DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE TOLD TO DO SO.**

## SECTION : NUMBER SYSTEM & DIVISIBILITY (123 QUESTIONS)

**1.** What is the remainder when 2n is divided by 4 ? What is the last digit in the expansion of [CDS (I) Paper 1 Q3]
(a) I only (b) 2 only (c) Both 1 and 2 (d) Neither I nor 2 What is x

**2.** If log, 6 m and log, 3 log, (r/2) equal to ? = n, then what is Consider the following statements : A number is divisible by 9 if a, + a2 + + + is divisible by [CDS (I) Paper 1 Q4]
(a) I only (b) 2 only (c) Both 1 and 2 (d) Neither I nor 2 What is x

**3.** a, - + a, - a, + as is divisible by [CDS (I) Paper 1 Q9]
(a) I only (b) 2 only (c) Both 1 and 2 (d) Neither I nor 2 What is x

**4.** What is the value Of 13th observation ? [CDS (I) Paper 1 Q39]
(a) 20 (b) 23 (c) 32 (d) 40 26

**5.** Consider those numbers between 300 and 400 such that when each number is divi&d by 6, 9 and 12. it leaves 4 as remainder in each case. What is the sum of the numbers ? [CDS (I) Paper 3 Q4]
(a) 692 (b) 764 (c) 1080 (d) 1092 - A 2

**6.** Which one of the following is a corect statement ? 13, What is the least integral value of k for which the equation has both the roots positive ? [CDS (I) Paper 3 Q12]
(a) 3 years (b) 4 years
(c) 5 years (d) 6 years 8. What is the median of the ages of minor children ?

**7.** If u,.v, w are real numbers such that u3 - 8v3 18uvw, which one of the following is conect ? (d) u +2v=-3w [CDS (I) Paper 3 Q10]
(a) 3 years (b) 4 years
(c) 5 years (d) 6 years 8. What is the median of the ages of minor children ?

**8.** What is the least integral value of k for which the equation has both the roots positive ? [CDS (I) Paper 3 Q13]
(a) 3 years (b) 4 years
(c) 5 years (d) 6 years 8. What is the median of the ages of minor children ?

**9.** If the highest common factor of two positive integers is 24 then their least Common multiple cannot [CDS (I) Paper 3 Q17]
(a) 72 (b) 216 (c) 372 (d) 600

**10.** For any integer n, what is the HCF Of integers m = + I and k 9n +4 ? [CDS (I) Paper 3 Q36]
(a) 4 days (b) 5 days (c) 6 days (d) days 33. If of

**11.** Which one of the following is an infinite set ? [CDS (I) Paper 3 Q38]
(a) {x: x is a whole number less than or equal to 1000} (b) {x: x is a natural number less than 1000}
(c) {x: x is a positive integer less than or equal to 1000} (d) {x: x is an integer and less than 1000}

**12.** A ladder of 17 ft. length reaches a window Which is 15 ft. above the ground on one side of the street. Keeping its foot at the same point the ladder is turned to the other side of the street and now it reaches a window 8 ft. high. What is the Width of the street ? [CDS (I) Paper 3 Q51]
(a) 23 ft (b) 15 ft. (c) 25 ft. (d) 30ft.

**13.** What is the weighted mean of first IO natural numbers whose weights are equal to the goffesponding numlYr ? (d) 4-5 [CDS (I) Paper 3 Q73]
(a) 4-5 74. What is volume of the frustum of a cone with height h and radii r, , ?
(b) 9th
(c) 15m
(d) I *5m

**14.** A hemispherical bowl of internal radius 20 cm contains sauce. The sauce is (o be filled in conical shaped bottles of radius 5 cm and height 8 cm. What is the number of bottles required ? [CDS (I) Paper 3 Q78]
(a) 100 (b) 90 (c) 80 (d) 75

**15.** If k is any even positive integer, then (k 2 + 2k) is [CDS (I) Paper 4 Q24]
(a) divisible by 24 (b) d ivisible 8 but may no t be divisible 24
(c) divisible by 4 but may not be divisible by 8 (d) divisible by 2 but may not be divisible by 4

**16.** I APB Cpl) i A, B, C [CDS (I) Paper 5 Q88]
(a) 121-7 121-75 (b) 121-75 ABC t, X Y AX.AYur (c) 16 2 (d) 20 cm 2

**17.** I APB Cpl) i A, B, C [CDS (I) Paper 5 Q84]
(a) 121-7 121-75 (b) 121-75 ABC t, X Y AX.AYur (c) 16 2 (d) 20 cm 2

**18.** Consider the following statements 64. [CDS (I) Paper 6 Q63]
(a) 36 (b) 44 (c) 52 64. (d) 150

*— SPACE FOR ROUGH WORK —*

**19.** Which one of the following is a prime [CDS (I) Paper 6 Q91]
(a) 280
(b) 52 What is the length of the perpendicular drawn from the centre Of circle of radius r On the chord of length r ?
(c) r/2 If a and ß are complementary angles, then sin a Cosa - i what is coseca * cosec ß sin cosß equal to ?
(d) None of the abo ve Consider the following statements : 88. 89. 90. 91. Which one of the following is a prime number ?

**20.** What is the sum Of their reciprocals ? [CDS (I) Paper 7 Q75]
(a) 64 minutes
(b) 80 minutes
(c) 96 minutes
(d) 120 minutes The pair of numbers which are relatively prime to each other is :

**21.** Consider the following Statements I. Every integer is a rational n umber. [CDS (I) Paper 8 Q11]
(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither I nor 2 The product of HCF and LCM of 18 and 15 is :

**22.** What is 0.242424.... [CDS (I) Paper 8 Q15]
(a) 23/99
(b) 8/33
(c) 7/33
(d) 47/198 equal to ? 4

**23.** Which one of the following has least number of divisors ? [CDS (I) Paper 8 Q18]
(a) 88
(b) 91
(c) 96
(d) 99

**24.** The price of an article is Rs. [CDS (I) Paper 8 Q25]
(a) x- 10
(b) x = 11
(c) x = 11.5 If 0.764 y = 1.236 x, Y - x value Of
(d) 0.764

**25.** What is the value of [CDS (I) Paper 8 Q27]
(a) x- 10
(b) x = 11
(c) x = 11.5 If 0.764 y = 1.236 x, Y - x value Of
(d) 0.764

**26.** If n is a prime number greater than 5, then n' - 1 is divisible by 2400. Every square number is of the form 5n or (5n - 1) or (5n + l) where n is a whole number. Which of the above Statements is/are correct ? [CDS (I) Paper 8 Q2]
(a) 1 only
(b) 2 only
(c) Both land 2
(d) Neither 1 nor 2 33. kx - 6, then what is the value Of

**27.** There are some benches in a class room having the number of rows 4 more than the number of columns. If each bench is seated with 5 students, there are two seats vacant in a class Of 158 students. The number of rows is (d) 10 [CDS (I) Paper 8 Q46]
(a) ac I If f
(b) -20
(c) 20
(d) 12

**28.** is divisible by [CDS (I) Paper 10 Q1]
(a) 15
(b) 25
(c) 20
(d) 30 9. 10. 11. What is equal to?

**29.** 64 74 48 (d) 42 [CDS (I) Paper 10 Q16]
(a) O only
(b) 20 only
(c) both 10 and 20
(d) neither IO nor 20 If

**30.** HOW many pairs Of positive integers m and n satisw the equation m n 12 where n is an odd integer less than 60? 17 (d) . The 7 4 3 sides Of a triangle are in the ratio - - : If its perimeter is 234 52 cm, then What is the length Of the smallest side? [CDS (I) Paper 12 Q16]
(a) 9 cm
(b) I O Crn
(c) 11 cm
(d) 12 cm

**31.** What is the remainder when x5 - 5x2 + 125 is divided by x +5? (b) 125 (c) -3125 (d) 3125 IA 16 *and ab(x2 - - b2)? (a2x2 bx)2 (a2x2 of 45 [CDS (I) Paper 12 Q48]
(a) 125
(b) -3125
(c) 3125 IA 46.
(d) e 160 only e 240 only e 160 or 240 Neither e 160 nor 240

**32.** Consider the following statements . [CDS (I) Paper 12 Q89]
(a) The triangle is isosceles
(b) The triangle is equilateral
(c) None Of the ab ove For the next three
(d) 3124

**33.** A runs I - times as fast as B. If A gives B a start of 80 m, how far If must the winning post from the starting point be so that A and B might reach it at the same time? [CDS (I) Paper 13 Q28]
(a) 200 m
(b) 300 m
(c) 270 m
(d) 160 m

**34.** Ihe number of values of x satisfying 100 x + - > 50 , where x is a natural number less than or equal to 100 is [CDS (I) Paper 14 Q1]
(a) 51
(b) 53
(c) 55
(d) 57

**35.** A number when divided by 7 leaves a remainder 3 and the resulting quotient when divided by 1 1 leaves a remainder [CDS (I) Paper 14 Q12]
(a) 35
(b) 40
(c) 45
(d) 50 106

*— SPACE FOR ROUGH WORK —*

**36.** If the same number when divided by 11 leaves a remainder m and the resulting quotient when divided by 7 leaves a remainder n. What are the values Of m and n respectively ? [CDS (I) Paper 14 Q6]

(a) I and 4 (b) 4 and 1 (c) 3 and 6 (d) 6 and 3 4

**37.** The seven digit .number 876p37q is divisible by [CDS (I) Paper 14 Q15]

(a) I only (b) 2 only

(c) I and 2 only l, 2 and Y is/are 14. If (d) 2 minutes

**38.** Which one of the following is correct ? [CDS (I) Paper 14 Q18]

(a) I only (b) 2 only

(c) I and 2 only l, 2 and Y is/are 14. If (d) 2 minutes

**39.** The number of pairs (x, y) where x, y are integers satisfying the equation 21x+48y=5is [CDS (I) Paper 14 Q21]

(a) Zero (b) One (c) Two (d) Infinity

**40.** Let x and y be positive integers such that x is prime and y is composite. Which of the following statements are correct ? 1. (y - x) can be an even integer. 2. can an even integer. 3. , can be an even integer. - A of Select the correct answer using the code given below : [CDS (I) Paper 14 Q22]

(a) I and 2 only (b) 2 and 3 only (c) I and 3 only (d) 1, 2 and 3

**41.** Let x and y be positive integers such that x > y. The expressions 3x 4 2}' and + 3y when divided by 5 leave remainders 2 and 3 respectively. What is the remainder when (x - y) is divided [CDS (I) Paper 14 Q23]

(a) 43-2%

(b) 56-8% 20. If k=x - y+2z where - 2<-r$l, - l y s 2 and 3 S z «*then which one of the following is correct ?

(c) 5<k$11 21. The number of pairs

(d) Zero

**42.** The sum of first 47 terms of the series 11 4564 56456 (b) 6 6 9 20 - A 12 34.. Which one of the fellowing is correct ? [CDS (I) Paper 14 Q33]

(a) (x + 2) is a factor of - + - 24.1 + 32 (b) (x +2) is a factor Of -12r2+24x-32

(c) (x - 2) is a factor Of x' + 32 (d) (x - 2) is a factor of x' + 0-1* +24x-32

**43.** A number consists of two digits, whose sum is [CDS (I) Paper 14 Q47]

(a) 100066-1 (b) 10000m

(c) 100+-6 (d) 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

**44.** Ihe product of digits of the number is (c) 10 (d) 12 [CDS (I) Paper 14 Q27]

(a) 100066-1 (b) 10000m

(c) 100+-6 (d) 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

**45.** What is their sum ? [CDS (I) Paper 14 Q3]

(a) 661 (b) 666 (c) 676 (d) 777

**46.** From a rectangular sheet of sides 18 cm and 14 cm, a semicircular portion with smaller side as diameter is taken out. Then the area of the remaining sheet will be [CDS (I) Paper 14 Q11]

(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

**47.** Consider the following statements in respect Of the expression where n is an integer : I. There are exactly two values of n for which Sn 861. and hence for any integer m, we have two values Of n for which Sn m. Which Of the above statements is/ are co rrect? [CDS (I) Paper 15 Q17]

(a) I only (b) 2 only (c) Both and 2 (d) Neither I nor 2

**48.** then which Of the following is/ are correct? Select the correct answer using the code given below. [CDS (I) Paper 15 Q19]

(a) I only (b) 2 only

(c) Both I and 2 (d) Neither I nor 2 710 - 510 is divisible by 10 (d) 11

**49.** Let a two-digit number be k times the sum of its digits. If thc number formed by interchanging the digits is m times the sum of the digits, then the value Of (b) 10-k (c) 11-k [CDS (I) Paper 15 Q21]

(a) I only (b) 2 only

(c) Both I and 2 (d) Neither I nor 2 710 - 510 is divisible by 10

**50.** Every natural number is a real number. Every real number is a rational number. Every integer is a real number. Every rational number is a real n umber. Which Of the above Statements are correc t? [CDS (I) Paper 15 Q4]

(a) I, 2 and 3 (b) 1, 3 and 4 (c) 2 and 3 0 nly (d) 3 and 4 only

**51.** For what value Of k is (x + I) a factor of (c) _2 Consider the following Statements : I. There exists a positive real number m such that cos x z 2m 2. mn m + n for all m, n belonging to set of natural numbers. Which of the above statements is/are correct? [CDS (I) Paper 15 Q35]

(a) I only (b) 2 only (c) Both I and 2 (d) Neither I nor 2

**52.** Consider thc following in respect of natural numbers a, b and c : [CDS (I) Paper 15 Q41]

(a) 38. Which of the points P (b) Q and R o nly (c) P and R o nly (d) P and Q o Illy

*— SPACE FOR ROUGH WORK —*

**53.** Which of the following is correct in respect Of the equation 3 e)? (Given that is a real number.) u13, - u13, (d) None of the above [CDS (I) Paper 15 Q68]

(a) 2 -Be , thenC+2B

(b) 0-1 0+1 cos26 - 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e

(c) only

(d) 2 only

**54.** If a and b are negative numbers and c i s a pos itive real num , then Which of the following is/ are c [CDS (I) Paper 15 Q99]

(a) AM < Median (b) GM Median

(c) AM = Me dian Suppose for Which one of the corre ct? (d) GM>1/6

**55.** If is an integer and a, are the roots of 4x2 - such that I < a < 2 4 and 2 < < 3, then how many values can take ? (c) 14 (d) 15 62 +72 +82 +92 +102 [CDS (I) Paper 16 Q2]

(a) x = 100 (b) x = IO (c) 330 (d) 340

**56.** A is a set of positive integers such that when divided by 2, 3, 4, 5 and the remainder I, 2, and 5 respec- tively. How many integers between 0 and 100 belong to the set A ? [CDS (I) Paper 16 Q6]

(a) No integer (b) One (c) Two (d) Three

**57.** "Ihe cost Of kg rice is Rs. [CDS (I) Paper 16 Q21]

(a) p -loniy 20. 6 If p and q are the roots of x2+px+q = 0, then which one of the following is correct ? remaining work ?

(b) 6 hours

(c) 12 hours

(d) hours

**58.** Which one Of the following is correct in respect of the number 1729 ? [CDS (I) Paper 16 Q22]

(a) It cannot be written as the sum Of the cubes of two positive integers

(b) It can be written as the Sum Of the cubes of two positive integers in one way only

(c) It can be written as the sum of the cubes Of two positive integers in two ways only

(d) It can be wHtten as the sum of the cubes Of two positive integers in three ways Only

**59.** Which one of the following statements is correct ? [CDS (I) Paper 16 Q3]

(a) Mahesh saves more than what Kamal saves (b) Savings of both of them are equal

(c) Kamal saves more than what Mahesh saves (d) It is not possible to determine who saves more

**60.** What is the value of tanl Otan20tan30tan40 . . tan890 ? [CDS (I) Paper 16 Q85]

(a) 150

(b) 300

(c) 450

(d) 600 82. Ifthe length of the shadow of a tower is equal to its height, then what is the Sun's altitude at that time ?

**61.** The average score of class X is [CDS (I) Paper 16 Q86]

(a) 150

(b) 300

(c) 450

(d) 600 82. Ifthe length of the shadow of a tower is equal to its height, then what is the Sun's altitude at that time ?

**62.** The average score of class Z is [CDS (I) Paper 16 Q76]

(a) 150

(b) 300

(c) 450

(d) 600 82. Ifthe length of the shadow of a tower is equal to its height, then what is the Sun's altitude at that time ?

**63.** If the average marks Of the complete class is 72, then what is the average marks Of the girls ? [CDS (I) Paper 16 Q75]

(a) 64 (b) 65 (c) 68 (d) 74

**64.** W'hat is the difference between the sum of the cubes and that Of squares Of first ten natural numbers ? [CDS (I) Paper 16 Q98]

(a) 2280 (b) 2640 (c) 3820 (d) 4130

**65.** The sum of the numbers is not divisible by(X+Y + Z). [CDS (I) Paper 19 Q1]

(a) ab 1

(b) abc equation 19. value of 20. zxr - 1, then aq ¯ ¯ P cP - q is equal to

(c) abc

(d) xyz

*— SPACE FOR ROUGH WORK —*

**66.** The sum of the numbers is divisible by [CDS (I) Paper 19 Q2]
(a) ab 1
(b) abc equation 19. value of 20. zxr - 1, then aq ¯ ¯ P cP - q is equal to
(c) abc
(d) xyz

**67.** Consider the following statements : [CDS (I) Paper 19 Q90]
(a) ab 1
(b) abc equation 19. value of 20. zxr - 1, then aq ¯ ¯ P cP - q is equal to
(c) abc
(d) xyz

**68.** What is åe of bikers in mad 2017 ? [CDS (I) Paper 21 Q9]
(a) 163 (b) 152 (c) 141 (d) 134

**69.** If of a Üid its rec9mcal is , thn numbers [CDS (I) Paper 21 Q71]
(a) 1 only (b) 96" cm2 (c) 2 only (d) Boå 1 2 69T cm2

**70.** If p divi&s qr if p divides q, p must divi& r. 2' 2' WNch of the [CDS (I) Paper 21 Q2]
(a) 1 only (b) 96" cm2 (c) 2 only (d) Boå 1 2 69T cm2

**71.** What is the numb' in 125, 823 md 920 ? [Givn hg102 = 0-301, log. - &477, 10867 - 0845] [CDS (I) Paper 21 Q77]
(a) 21, 20, 19
(b) 20, 19, 18
(c) 22, 21,20
(d) n, 20,21 LA x be integer such that 14 divid« x, is 7; wha 15 x, is

**72.** The number 2x3*5x7x11+1 is [CDS (I) Paper 22 Q1]
(a) a prime number (b) not a prime, but power of a prime
(c) not a power of a prime, but a composite even number (d) not a of a prime, but a composite odd number

**73.** If the sum of the digits of a number 10" - 1, where n is a natural number, is equal to 3798, then what is the value of n? [CDS (I) Paper 22 Q6]
(a) 421 (b) 422 (c) 423 (d) 424

**74.** If the points p and Q represent real numbers 0-73 and 0-56 on the number line, then what is the distance between P and Q ? 6 16 45 11 90 9 [CDS (I) Paper 22 Q25]
(a) 49x2 -f- 3x + 2 (b) 49x2 _ 3x + 2
(c) 49++2 (d) 59++2 24. If x varies asyz, theny varies inversely

**75.** What is the number of students in the class ? [CDS (I) Paper 22 Q5]
(a) 100 (b) 105 (c) 110 (d) 120

**76.** The number of three digit numbers (all digits are different) which are divisible by 7 and also divisible by 7 on reversing the order Of the digits, is [CDS (I) Paper 22 Q41]
(a) Six (b) Five (c) Four (d) Three

**77.** S is always divisible by 3 and divisible by 9 divisible by 37 15 15 Select the corect answer using the code given below : [CDS (I) Paper 22 Q3]
(a) 1 only (b) 2 only (c) 1 and 2 (d) I and 3

**78.** What is the least perfect square which is divisible by 3, 4, 5, 6 and 7? [CDS (I) Paper 22 Q47]
(a) 1764 (b) 17640 (c) 44100 (d) 176400

**79.** Consider the following statements with reference to the given figure : [CDS (I) Paper 22 Q100]
(a) 924 square units
(b) 824 square units
(c) 624 square units
(d) 462 square units 99. 6 cm What is the approximate area of the shaded region in the figure given ?

**80.** The sum of the areas of AAOD and ABOC is equal to the sum of the areas of MOB and ADOC [CDS (I) Paper 22 Q1]
(a) 924 square units
(b) 824 square units
(c) 624 square units
(d) 462 square units 99. 6 cm What is the approximate area of the shaded region in the figure given ?

**81.** For how many real values of k is 2 + 12kx 16 a "uare for every integer x ? [CDS (I) Paper 24 Q21]
(a) Zem (b) One (c) Two (d) Four

*— SPACE FOR ROUGH WORK —*

**82.** If equal to? 195 16 16 then What is If the equation 4x 2 - 2kx + 3k O has equal roots, then What are the values of k? [CDS (I) Paper 24 Q22]
(a) Zem
(b) One
(c) Two
(d) Four 22. If equal to? 195 16 16 4x3 + 12x2 - x - 3 is divisible by

**83.** For how many real values of k is 2 + 12kx 16 a "uare for every integer x ? [CDS (I) Paper 24 Q24]
(a) Zem (b) One (c) Two (d) Four

**84.** 60* 45' B t, cote [CDS (I) Paper 24 Q60]
(a) 11 (b) 22
(c) 77 (d) 99 k sece 59. 60' 60. 60* 45'

**85.** Later on 5 grace marks are added to each student. What are the average marks of the revised marks Of the students? 30 31 32 1 P.T.O. [CDS (I) Paper 24 Q32]
(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015?
(b) and E
(c) A, C and D only
(d) C D and E

**86.** What are distinct prime mctors of the number 26381 ? [CDS (I) Paper 25 Q1]
(a) 29, 17, 37 (b) 31, 17, 47 (c) 19, 37, 13 (d) 23, 31, 37

**87.** Which one of the following is a factor of the [CDS (I) Paper 25 Q2]
(a) 29, 17, 37 (b) 31, 17, 47
(c) 19, 37, 13 (d) 23, 31, 37 2. Which one of the following is a factor of the

**88.** What is the square root of 23 - 4*v'fiS ? (b) 7-3-6 [CDS (I) Paper 25 Q3]
(a) 29, 17, 37 (b) 31, 17, 47
(c) 19, 37, 13 (d) 23, 31, 37 2. Which one of the following is a factor of the

**89.** If x is a negative real number, then which of the following are not correct ? I. There is some natural number k such that > 0 2. + x > O always 4. x2 is always a rational number Select the correct answer using the code given below : [CDS (I) Paper 25 Q7]
(a) 1, 2 and 3 (b) 1, 2 and 4 (c) and 4 (d) 2, 3 and 4

**90.** What is (d) [CDS (I) Paper 25 Q23]
(a) 30 (b) 45 (c) 60 (d) 75 equal to ?

**91.** A two-digit number is such that the product of the digits is [CDS (I) Paper 25 Q32]
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2 only when Ware correct ? 28. Let work done by

**92.** If 63 is added to this number, the digits interchange their places. What is the sum of the digits in the number ? [CDS (I) Paper 25 Q8]
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2 only when Ware correct ? 28. Let work done by

**93.** What is Z BAP equal to ? [CDS (I) Paper 25 Q89]
(a) 300 (b) 400 (c) 450 (d) 600

**94.** How many such sequences of five consecutive integers are possible ? [CDS (I) Paper 25 Q99]
(a) One (b) Two (c) Three (d) Five

**95.** - land 2 1-1? [CDS (I) Paper 28 Q5]
(a) 34
(b) 90
(c) 127
(d) 129 What is the largest power Of IO that divides the product 29 x 28 x 27 x 2 x I ?

**96.** Question : Is p2 + q2 + q Odd, where p , q are positive integers ? [CDS (I) Paper 28 Q78]
(a) 24 (b) 36 (c) 48 (d) 60

**97.** Question : What is the cost of 15 pens, 21 pencils and 18 note books ? The cost of 7 pens, 6 pencils and 5 note books is [CDS (I) Paper 29 Q99]
(a) 15° (b) 30° (c) 45° (d) 60°

**98.** Let N be a 5-digit number. When N is divided by 6, 12, 15, 24 it leaves respectively 2, 8, 11, 20 as remainders. What is the greatest value of N ? [CDS (I) Paper 31 Q3]
(a) 99960 (b) 99956 (c) 99950 (d) 99946

*— SPACE FOR ROUGH WORK —*

**99.** What is the remainder when 111222 + 222333 + 333444 is divided by 5 ? -3 S. What are the last digits in the multiplication of 4321012345 x 98766789 ? 6. p varies directly as (x-2 + Y2 + zz). When 3, thenp - 70. What is the value of p when x - 1, y = 1, [CDS (I) Paper 31 Q4]

(a) 100 (b) 125 (c) 135 (d) 140

**100.** What is the sum of the largest and the smallest 4-digit numbers made by using single digit prime numbers (without repetition) ? [CDS (I) Paper 31 Q24]

(a) 7887 (b) 7997 (c) 8998 (d) 9889

**101.** What is the remainder when 3255 is divided by 28 ? (b) 11 (c) 24 (d) 27 [CDS (I) Paper 31 Q25]

(a) nM - k (b) n (c) 81 (d) 105

**102.** What is the value of x 8) if (10097 + 10054 + x + 1) leaves a remainder O when divided by 9 ? A - [CDS (I) Paper 31 Q26]

(a) nM - k (b) n (c) 81 (d) 105

**103.** Consider the following statements : I. The line segment AP divides the area of the triangle ABC into two equal parts [CDS (I) Paper 31 Q53]

(a) I and 11 only (b) 11 and Ill only

(c) I and Ill only (d) 1, 11 and 111 Consider the following for the next two

**104.** What is OQ equal to ? [CDS (I) Paper 31 Q58]

(a) 5 cm (b) 6 cm 59. What is PM equal to ?

(c) IO cm (d) 12 cm

**105.** What is PM equal to ? [CDS (I) Paper 31 Q59]

(a) 5 cm (b) 6 cm 59. What is PM equal to ?

(c) IO cm (d) 12 cm

**106.** divides 675 + 335 Which of the statements given above is/ are correct ? [CDS (I) Paper 31 Q11]

(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

**107.** Consider the following statements in respect of where n is a natural number : I. p is always (Xid 11. p is a perfect square Which of the statements given above is/ are correct ? [CDS (I) Paper 31 Q79]

(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11 A -

**108.** If + px2 + qx + r is an integer for all integral values of x, then consider the following statements : I. p must be an integer 11. q must be an integer Ill. r must be an integer Which of the statements given above is/are correct ? [CDS (I) Paper 31 Q88]

(a) I and 11 only (b) Ill only

(c) I, IIandIll (d) None of the statements is correct A -

**109.** The number 199 can be written as m2 - n2, where m, n are natural numbers (m > n). What is the value of mn ? [CDS (I) Paper 31 Q96]

(a) 9900 (b) 9800

(c) 9701 (d) Cannot be uniquely determined

**110.** Let S - 7b + IIC +13d, where a, b, c and d are natural numbers. What is the number of distinct remainders of S when it is divided by 10 ? (d) More than 5 [CDS (I) Paper 31 Q99]

(a) 1100

(b) 1200

(c) 1300

(d) 1400 96. The number 199 can be written as m2 - n2, where m, n are natural numbers

**111.** Let p and q be two natural numbers such that (p + is divisible by [CDS (I) Paper 33 Q20]

(a) 12 21. What is c2 + bc - a2 - ab equal to?

(b) -1/2 23. 24. 25. If x -2+21/2 +23/2 x2 - 4x - IO equal to? If equal to ?

(c) 2p

(d) then what is then what is x 22. If ab ba , then what is equal to?

**112.** If 4, Where O then what is cosece equal to? (b) 3/2 5/3 [CDS (I) Paper 33 Q32]

(a) 3/2 5/3 33. If 2 tane=sec29 - 2, where R 12, then what is cote equal to?

(b) v'ä-I

(c) 0+2 I - sin 34. What is

(d) mane

**113.** Which Of the following is/ are the factor(s) of(3x+ y) 2 + (3x + 5y) + 5y)2? [CDS (I) Paper 33 Q39]

(a) 11 only (b) Both I and 11

(c) Neither I nor 11 39. Which Of the is/ are the Of (d) Both I and 11

**114.** What is tanß equal to? p.T.O. [CDS (I) Paper 33 Q41]

(a) 11 only (b) Both I and 11

(c) Neither I nor 11 39. Which Of the is/ are the Of (d) Both I and 11

*— SPACE FOR ROUGH WORK —*

**115.** What is Wc e equal to? (c) 1 P.T.O. [CDS (I) Paper 33 Q47]
(a) mn - I O For the following two
(b) 2/3 For the following tvo
(c) 1 P.T.O. For the following two
(d) 2/3 sina 4 u sin # cosa cosß 9 45. What is tan 2 equal to? 46. What is tan 2 ß equal to? 1/3 For the following tvo

**116.** Let N be the greatest number that will divide 600, 631 and 724, leaving the same remainder. What is the value of N? [CDS (I) Paper 34 Q24]
(a) 41 (b) 37 (c) 33 (d) 31 11

**117.** Consider the following statements : I. Two consecutive natural numbers are always co-prime. [CDS (I) Paper 34 Q28]
(a) I only (b) 11 only
(c) Both land 11 (d) Neither I nor 11 29. If and then what is log10

**118.** Let P = N2 where N is an odd integer. What is the remainder when P is divided by 8 ? (d) Cannot be determined due to insufficient data [CDS (I) Paper 34 Q31]
(a) I only (b) 11 only
(c) Both land 11 (d) Neither I nor 11 29. If and then what is log10

**119.** The minute hand of a clock is 15 cm long and sweeps an area of 15" cm on the dial of the clock. How much angle does it describe during this period ? [CDS (I) Paper 34 Q60]
(a) 120 (b) 180 (c) 240 (d) 300

**120.** What is 25 16 16 19 4 4 [CDS (I) Paper 34 Q83]
(a) 487 (b) 483 (c) 481 (d) 480 85. What

**121.** What (X2 - Y2) equal to ? (d) 8800 [CDS (I) Paper 34 Q85]
(a) 487 (b) 483 (c) 481 (d) 480 85. What

**122.** What is tang equal to ? (C) pq [CDS (I) Paper 34 Q93]
(a) 15° (b) 30° (c) 45° (d) 60°

**123.** What is tan2a equal to ? (b) [CDS (I) Paper 34 Q95]
(a) 96. What is cot2ß equal to ? p For the next two (b) 10
(c) 13 For the next two (d) 2pq - -

## SECTION : HCF, LCM & LOGARITHMS (23 QUESTIONS)

**124.** What *is the HCF 4x3 + 3x2y - 9xy2 uy3 [CDS (I) Paper 4 Q7]
(a) x - 2y of and third side in cm? (b) g, 2v'fi)
(c) (442 (d) (44ä, 21,110)

**125.** Rs. 240 Rs. 480 Rs. 800 are in the ratio of 12 : 23. second number ? [CDS (I) Paper 6 Q4]
(a) 44
(b) 55
(c) 66
(d) 77 What is - l? the HCF x3 _ of the 3x2 + (d) None of the above -7Y+10 3x What is the polynomials 1 and Ilx +4y - 10 9

**126.** Which one of the following is a prime number ? [CDS (I) Paper 6 Q90]
(a) 161
(b) 171
(c) 173
(d) 221 What is (a) 16 equal to ? What is log,o - + log,o ¯ + log.o - + up to 8 terms equal to ? (a) (c) (d)

**127.** equal to ? [CDS (I) Paper 7 Q41]
(a) 5/4 (b) 6/5
(c) 25/31 (d) 31/42 The sum of two numbers is 232 and their HCF is

**128.** If 18 is added to the number, the digits are reversed. The number is equal to [CDS (I) Paper 8 Q8]
(a) 26
(b) 35
(c) 53
(d) 62 If log, o a = p and loglo b = q, then what is loglo (aP bq) equal to ? (a) P2 + q2 (c) p2q2 q

**129.** If (x+l) is the HCF of Ax2+Bx+C and Bx2 + Ax + C where A B, then the value of C is [CDS (I) Paper 14 Q14]
(a) I only (b) 2 only
(c) I and 2 only l, 2 and Y is/are 14. If (d) 2 minutes

*— SPACE FOR ROUGH WORK —*

**130.** The LCM of two numbers is 12 times their HCF. The sum Of HCF and LCM is [CDS (I) Paper 14 Q86]
(a) 20 square t
(b) 30 square unit
(c) 40 square unit
(d) None of the above 84. Chord CD intersects the diameter AB of a circle at riW1t angle at a m

**131.** LCM(ab, = a LCM(b, c) [CDS (I) Paper 15 Q1]
(a) 38. Which of the points P (b) Q and R o nly (c) P and R o nly (d) P and Q o Illy

**132.** HCF(ab, HCF(b, c) [CDS (I) Paper 15 Q2]
(a) 38. Which of the points P (b) Q and R o nly (c) P and R o nly (d) P and Q o Illy

**133.** HCF(a, b) < LCM(a, b) [CDS (I) Paper 15 Q3]
(a) 38. Which of the points P (b) Q and R o nly (c) P and R o nly (d) P and Q o Illy

**134.** HCF(a, b) divides LCM (a, b). Which of the above are correct? [CDS (I) Paper 15 Q4]
(a) I and 2 Only (b) 3 and 4 only (c) , 2 and 4 only (d) 1, 2, 3 and 4

**135.** Wych me Of xver LCM ? [CDS (I) Paper 21 Q12]
(a) be 2 one Of the 61%wing ? 13. HCF numbes is 12. Wych me Of xver LCM ?
(b) 80
(c) 60 36 *
(d) 24 14. WEch Of

**136.** What is value of bgIdsin9) + log10(tnØ) + log I o(cote) + loglßw9) + ? (c) 0-5 [CDS (I) Paper 21 Q31]
(a) Its lia ixuide O
(b) Its lies on the triülgle
(c) It 30. Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ?
(d) AB<RX

**137.** How many digits are there in (54)10 ? (Given that log102 - 0-301 and log,03 0477) [CDS (I) Paper 22 Q3]
(a) 16 (b) 18 (c) 19 (d) 27 3 (b) (c) 3

**138.** What is the HCF of x 3 - 30 and -5x+6? (x +41x-3) (c) (d) [CDS (I) Paper 24 Q16]
(a) Which Of the above statements is/are correct? (b) I only
(c) 2 only (d) Both I and 2

**139.** A circle touches all the four sides of a quadrilateral ABCD. If AB - 9 cm, BC 8 cm and CD - 12 cm, then what is DA equal to? [CDS (I) Paper 24 Q90]
(a) 14 cm (b) 13 cm (c) 12 cm (d) I lcm

**140.** What is the remainder aner dividing the number by 9 ? S. The sum of LCM and HCF of two numbers is 536 and the difference between LCM and HCF is [CDS (I) Paper 25 Q4]
(a) 29, 17, 37 (b) 31, 17, 47
(c) 19, 37, 13 (d) 23, 31, 37 2. Which one of the following is a factor of the

**141.** If loglox+logIox2 = 2 log,ox+I, then what is the value ofx ? (d) 10 5 5 [CDS (I) Paper 25 Q13]
(a) only (b) k>2v5 only
(c) k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is (d) I - aßy

**142.** Suppose p and q are the LCM and HCF respectively of two positive numbers. If p : q = 14 I and pq 1134, then what is the difference between the two numbers? [CDS (I) Paper 26 Q5]
(a) 27 (b) 35
(c) 45 (d) Cannot be determined due insufficient data

**143.** What is the HCF of 236-1 and 245-1 ? [CDS (I) Paper 31 Q91]
(a) 1023 (b) 512 (c) 511 (d) 255

**144.** What is the remainder when (1725 1925) is divided by 18? (c) . The 3 9 HCF of x and y is H. Consider 12 13. the following statements in respect of the HCF of p - [CDS (I) Paper 33 Q11]
(a) 106
(b) 107
(c) 108 What is the remainder when x6 divided by x 2 + ? is 11. What is the remainder when
(d) 0 9 HCF of x and y is H. Consider

**145.** If (x - 5) is the HCF of - x - p and x2 - qx - IO, then what is the value of (P+q)? [CDS (I) Paper 34 Q6]
(a) -23 (b) -17 (c) 17 (d) 23 A -

*— SPACE FOR ROUGH WORK —*

**146.** HCF ofp, (p-q) is G Select the answer using the code given [CDS (I) Paper 34 Q11]
(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

## SECTION : PERCENTAGES, PROFIT, LOSS & DISCOUNT (33 QUESTIONS)

**147.** If the height of a cone*is increased by 50%, then what is ihe percentage increase in the volunie Of the cone? [CDS (I) Paper 4 Q71]
(a) 100/3 (b) 40 (c) 50 (d) 200/3

**148.** If 28 [CDS (I) Paper 4 Q93]
(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square
(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87* If

**149.** ABCD is a quadrilateral, the sides of Which touch a circle. Which one Of the following is correct? [CDS (I) Paper 4 Q92]
(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square
(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87* If

**150.** D 3k E BC 3 ADE 3 cm2 I EC ? [CDS (I) Paper 5 Q92]
(a) 16 cm (b) 20 em
(c) 25 cm (d) 30 cm ABC !3T3-å AB AC qi D 3k E BC 3 ADE 3 cm2 I EC ?

**151.** 16Jä -8T 16Jä - 27 -A) [CDS (I) Paper 5 Q95]
(a) 16 cm
(b) 20 em
(c) 25 cm
(d) 30 cm ABC !3T3-å AB AC qi D 3k E BC 3 ADE 3 cm2 I EC ? (a) 1-5 C m (d) 2-1 cm qx 9,680 10,648 I (c) 15%

**152.** Two students A and B solve an equation Of the form ± px + q = O. A starts with a %T0ng value of p and obtains the roots as 2 and [CDS (I) Paper 7 Q63]
(a) p (b) 3 and - 4 (c) - 3 and - 4 (d) - 3 and 4

**153.** What is 5% of of 500 ? [CDS (I) Paper 8 Q17]
(a) 12.5 (b) 25 (c) 1.25 (d) 6.25

**154.** A man buys a television set which lists for Rs. 5,000 at 10% discount. He gets an additional 2% discount (after the first discount) for paying cash. What does he actually pay for the set ? [CDS (I) Paper 8 Q19]
(a) Rs. 4,410 (b) Rs. 4,400 (c) Rs. 4,000 (d) Rs. 4,500

**155.** -Ai if one Of the Of the - bx + c O the square at the then Which one the following is ? 3bc c c Sbe + b [d) 3be 4 b3 + then what is x What ig the value Of O ne Ot factors tf - 5x2 - 81 ia ta) equgi to ? Let A : % is odd integer) and B - : xa - 8x + % - [CDS (I) Paper 9 Q22]
(a) 5% (b) 8% (c) 10% (d) 12.5%

**156.** 1%en which one of the ig ? (b) (41 ACB sum Of numbers 7 and sum Of 20. their ' is [CDS (I) Paper 9 Q1]
(a) Rs. 1,200 (b) Rs. 1,440 (c) Rs. 1,500 (d) Rs. 1,800

**157.** A person sold an article for e 3,600 and got a profit of 20%. Had he sold the article for 3,150, how much profit would he have got? [CDS (I) Paper 10 Q8]
(a) 15 (b) 25
(c) 20 (d) 30 9. 10. 11. What is equal to?

**158.** A person's salary has increased from e 7,200 toe 8,100. What is the percentage increase in his salary? 23. *Nyo 6 (c) 164% (d) 124% [CDS (I) Paper 10 Q21]
(a) O only (b) 20 only (c) both 10 and 20 (d) neither IO nor 20 If

**159.** A person selling an article for 96 finds that his loss percent is one-fourth Of the amount Of rupees that he paid for the article. What can be the cost price? (d) e 160 only e 240 only e 160 or 240 Neither e 160 nor 240 If (x + k) is the common factor of x + b and x2 + d, then What is k equal to? [CDS (I) Paper 12 Q46]
(a) 125 (b) -3125
(c) 3125 IA 46. (d) e 160 only e 240 only e 160 or 240 Neither e 160 nor 240

**160.** A person selling an article for 96 finds that his loss percent is one-fourth Of the amount Of rupees that he paid for the article. What can be the cost price? [CDS (I) Paper 12 Q47]
(a) 125 (b) -3125
(c) 3125 IA 46. (d) e 160 only e 240 only e 160 or 240 Neither e 160 nor 240

*— SPACE FOR ROUGH WORK —*

**161.** The value of a single discount on some amount which is equivalent to a series of discounts of 10%, 20% and on the same amount, is equal to [CDS (I) Paper 14 Q19]

(a) 43-2%

(b) 56-8% 20. If k=x - y+2z where - 2<-r$l, - l y s 2 and 3 S z «*then which one of the following is correct ?

(c) 5<k$11 21. The number of pairs

(d) Zero

**162.** A's salary was increased by and then decreased by 20%. On the whole A's salary is increased by [CDS (I) Paper 14 Q46]

(a) 100066-1 (b) 10000m

(c) 100+-6 (d) 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

**163.** In an election 10% of the voters on the voter list did not cast their vote and 60 voters cast their ballot papers blank. nere were Only two candidates. The winner was supported by 47% of total voters in the voter list and he got 308 votes more than his rival. The number of voters on the voter list is [CDS (I) Paper 14 Q49]

(a) 3600 (b) 6200 (c) 6028 (d) 6400

**164.** ABCD is a square. If the sides AB and CD are increased by 30%, sides BC and AD are increased by 20%, then the area of the resulting rectangle exceeds the area of the square by (b) 52% (c) 540/0 (d) 56% [CDS (I) Paper 14 Q72]

(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

**165.** Le g nd b be real + = 32 50 + b) 7 (d) be 2 one Of the 61%wing ? [CDS (I) Paper 21 Q11]

(a) be 2 one Of the 61%wing ? 13. HCF numbes is 12. Wych me Of xver LCM ?

(b) 80

(c) 60 36 *

(d) 24 14. WEch Of

**166.** The &iin nd bus two is in the ratio 3 4, If the by md bus ftre incras - by 30%, is åe mtio reviEd bus ? 9 [CDS (I) Paper 21 Q23]

(a) 17 12 32 19 21 24. N is by 17, is equal to 182. ne b«wen the quotiai md is 175. is åe value of N ?

(b) 2975

(c) 3094

(d) 3101

**167.** The price of an article X increases by 20% every year and price of article Y increases by 10% every year. In the year 2010, the price of article X was C 5000 and price Of article Y was 0000. In which year the difference in their prices exceeded 0000 for the first time ? [CDS (I) Paper 22 Q21]

(a) 2012 (b) 2013 (c) 2014 (d) 2015

**168.** Jails with highest and smallest percent- age of trained convicts are respectively [CDS (I) Paper 22 Q38]

(a) F and D (b) D and F (c) C and A (d) D and A

**169.** If the annual income of X is 20% more than that of then the income of Y is less than that of X by pO/o. What is the value of p ? [CDS (I) Paper 22 Q46]

(a) 10 (b) 16i (c) 36 (d) 27

**170.** A EErson bought a chair and a table for *750. He sold the chair at a gain of 5% and the table at a gain of 20%. He gained 16% on the whole. What is original cost of table ? [CDS (I) Paper 25 Q43]

(a) *400 (b) *450 (c) V550 (d) e600

**171.** In which one of the following cate- gories of employees, the percentage change in number of employees in 2020 was maximum as compared to 2019 ? [CDS (I) Paper 25 Q92]

(a) 101 *39 cm2 (b) 111-39cm2

(c) 121 *39cm2 (d) 131-39cm2 Consider the following for the next two

**172.** What was the percentage increase of category-E employees in 2020 as compared to that in 2019 ? [CDS (I) Paper 25 Q93]

(a) 101 *39 cm2 (b) 111-39cm2

(c) 121 *39cm2 (d) 131-39cm2 Consider the following for the next two

**173.** There are two categories of employees whose strength increased by same percentage in 2020. What are the two categories ? [CDS (I) Paper 25 Q94]

(a) E and F (b) C and E (c) A and E (d) and C A -

**174.** In which year was the percentage in- crease in production of cars minimum as compared to its previous year ? 27 [CDS (I) Paper 25 Q97]

(a) 2015 (b) 2016 (c) 2017 (d) 2018

**175.** There are two employees X and Y. X's salary is first increased by 12% and then decreased by 10%, and Y's salary is first increased by 10% and then decreased by 12%. If their salaries at present are equal, then what was the ratio of initial salary of X to initial salary of Y? [CDS (I) Paper 33 Q53]

(a) 50: 53 (b) 51: 53 (c) 121: 126 (d) 121: 125 I P.T.O.

*— SPACE FOR ROUGH WORK —*

**176.** A shopkeeper gives three consecutive discounts 10%, and 25% after which he sells the article at a profit Of 8% on the cost price. Had he sold the article after the first discount, how much profit would he have got? [CDS (I) Paper 33 Q60]

(a) 20%

(b) 40%

(c) None of the above 61. Let AD be the altitude of a triangle ABC If and

(d) I only

**177.** A reduction of 10% in the price of sugar enables a person to buy kg more for *2790. What is the reduced price per kilogram ? (b) z48 [CDS (I) Paper 34 Q32]

(a) I only (b) 11 only

(c) Both land 11 (d) Neither I nor 11 29. If and then what is log10

**178.** What is the percentage of copper in (b) 64% [CDS (I) Paper 34 Q87]

(a) 487 (b) 483 (c) 481 (d) 480 85. What

**179.** What is the percentage of zinc in X ? [CDS (I) Paper 34 Q88]

(a) 400/0 (b) 36% (c) 34% (d) 30% A -

## SECTION : SIMPLE & COMPOUND INTEREST, RATIO & MIXTURES (22 QUESTIONS)

**180.** The wages, of labourers in a factory has increased in the ratio 22 : 25 and their number decreased in the ratio 3 : [CDS (I) Paper 3 Q34]

(a) 4 days (b) 5 days (c) 6 days (d) days 33. If of

**181.** If two corresponding sides Of two similar triangles, are in the ratio 9 : 4, then What * the ratio Of their areas ? (c) 81 16 / 26A (d' 27 : 8 28 [CDS (I) Paper 4 Q89]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87* If

**182.** The ratio of two complementary angles is I : [CDS (I) Paper 7 Q91]

(a) 64 minutes

(b) 80 minutes

(c) 96 minutes

(d) 120 minutes The pair of numbers which are relatively prime to each other is :

**183.** The ratio of two complementary angles is I : [CDS (I) Paper 7 Q90]

(a) 64 minutes

(b) 80 minutes

(c) 96 minutes

(d) 120 minutes The pair of numbers which are relatively prime to each other is :

**184.** Sex ratio is defined as the number of females per 1000 males. In a place the total inhabitants are out of which 935,000 are females. What is the sex ratio for the place ? [CDS (I) Paper 8 Q20]

(a) 935 (b) 1000 (c) 1935 (d) 9350 6 2 23 2

**185.** The speeds of A, B and C are in the ratio 50 : 45 : [CDS (I) Paper 14 Q2]

(a) 43-2%

(b) 56-8% 20. If k=x - y+2z where - 2<-r$l, - l y s 2 and 3 S z «*then which one of the following is correct ?

(c) 5<k$11 21. The number of pairs

(d) Zero

**186.** A water tank, open at the top, is hemi- spherical at the bottom and cylindrical above it. Ihe radius is 12m and the capacity is 3312rt m3. The ratio of the surface areas of the spherical and cylindrical portions is - A [CDS (I) Paper 14 Q73]

(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

**187.** The area of a trapezium is 336cm2. If its parallel sides are in the ratio 5 : 7 and the perpendicular distance between them is 14 cm, then the smaller of the parallel sides is [CDS (I) Paper 14 Q78]

(a) 20 cm (b) 22 cm (c) 24 cm (d) 26 cm

**188.** A sum of Rs. 10,000 is deposited for 1 year at the rate of interest 10% com- pounded half yearly. What will be the interest at the end of one year ? [CDS (I) Paper 14 Q88]

(a) Rs. 1000 (b) RK1025

(c) Rs. 1050 (d) RS. 1100 a mixture of milk and water of

**189.** The quantity of water to be added to the mixture to make the * ratio of milk and water 1 : 2 is [CDS (I) Paper 14 Q3]

(a) 30 (b) 32 (c) 33 (d) 35

*— SPACE FOR ROUGH WORK —*

**190.** TWO men, and B run a 4 km racc on a course km round. If their speeds are in the ratio 5 : 4, how often does the winner pass the other ? [CDS (I) Paper 16 Q12]

(a) Once (b) Twice (c) Thrice (d) Four times - A 4

**191.** If a = xyp - l, b c aq ¯ ¯ P cP - q is equal to = zxr - 1, then [CDS (I) Paper 19 Q22]

(a) ab 1

(b) abc equation 19. value of 20. zxr - 1, then aq ¯ ¯ P cP - q is equal to

(c) abc

(d) xyz

**192.** Two cyli&s of have in ratio 2 : [CDS (I) Paper 21 Q49]

(a) 32%

(b) 30% Sl. If the Of h+ue of a trian* is, IO is tie of awh a riørt mgld ?

(c) cm2

(d) 50

**193.** The ratio of speeds of Xand Y is 5 : [CDS (I) Paper 22 Q28]

(a) 49x2 -f- 3x + 2 (b) 49x2 _ 3x + 2

(c) 49++2 (d) 59++2 24. If x varies asyz, theny varies inversely

**194.** The length and breadth of a rectangle are in the ratio 4 : [CDS (I) Paper 22 Q76]

(a) - *E (b) litre (c) litre , (d) - litre 10

**195.** If x is directly proportional to z and y is directly proportional to z, then (x 2 - y2) is directly proportional to za. If x is inversely proportional to z and y is inversely proportional to z, then (xy) is inverseb' Which Of the above statements is/are correct? [CDS (I) Paper 24 Q2]

(a) I only (b) 2 only (c) Both I and 2 (d) Neither 1 nor 2

**196.** A sum of money compounded annually doubles itself in 5 years. In how many years will it become four times of itself? [CDS (I) Paper 25 Q37]

(a) IO years (b) 12 years (c) 15 years (d) 20 years 11 11

**197.** 10% annum on compound interest for 3 years, compounded annually, and paid back in 3 equal annual installments during these years. What is the amount Of each installment (approximately) ? [CDS (I) Paper 33 Q58]

(a) e 4,437 (b) e 4,237 (c) r 4,021 (d) r 3,811

**198.** In what duration of time will *3300 become *3399 at 12% per annum interest compounded quarterly ? [CDS (I) Paper 34 Q12]

(a) 3 months (b) 6 months (c) 9 months (d) 1 year

**199.** What is the initial rate of interest per annum ? [CDS (I) Paper 34 Q89]

(a) Rs. 1,200 (b) Rs. 1,440 (c) Rs. 1,500 (d) Rs. 1,800

**200.** What is the ratio of the interest on the first loan to the interest on the second loan ? For the next two (02) items that follow : Let 12(tane + cot0) = 25, where 450 < 0 < [CDS (I) Paper 34 Q90]

(a) 1/2 (b) 1/√2 (c) √3/2 (d) 1

**201.** What is p4q2 + p2q4 equal to ? A - For the next two (02) items that follow : A person borrows a sum Of e 16,000 on simple interest at the beginning of a year. Aner 4 months, *24,000 more is borrowed at a rate of interest double the previous one. At the end of the year, the total interest on both the loans is *4800. [CDS (I) Paper 34 Q94]

(a) 4 : 5 (b) 5 : 6 (c) 3 : 4 (d) 2 : 3

## SECTION : TIME, SPEED & DISTANCE, TIME & WORK (43 QUESTIONS)

**202.** If 6 men and 8 boys can do a piece of work in 10 days while 26 men and 48 boys can do the sarru in 2 days, what is the time taken by 15 men and 20 boys in doing the same type of work ? [CDS (I) Paper 3 Q32]

(a) 4 days (b) 5 days (c) 6 days (d) 7 days

**203.** The inner and outer radii of a 7 m long hollow iron right circular cylindrical pipe are 2 cm and 4 cm respectively. If 1000 cni3 of iron weighs 5 kg, what is the weight of the pipe ? [CDS (I) Paper 3 Q86]

(a) 264 kg (b) 132 kg (c) 396 kg (d) None of the above

**204.** What is, the mean proportional between and (27 ? (b) 146 A stock Of food is enough for 240 men for 48 days. How long will the same stock last for 160 men ? [CDS (I) Paper 7 Q33]

(a) 54 days

(b) 60 days

(c) 64 days

(d) 72 days 9 2 3 = 2 and If x+Y x+y x - Y then what is ¯ equal to ? (a) 3/2 (c) 2/3 (d) 1/5 4

**205.** A car is traveling at a constant rate of 45 km per hour. The distance travelled by car from 10.40 am to 1.00 pm is [CDS (I) Paper 8 Q23]

(a) 165 km (b) 150 km

(c) 120 km (d) 105 km A number consists of two digits whose sum is

*— SPACE FOR ROUGH WORK —*

**206.** Three planets revolve round the sun once in 200, 250, days respectively in their own orbits. When do they all come relatively to the same position as at a certain point of time in their orbits ? [CDS (I) Paper 8 Q28]

(a) After 3000 days (b) After 2000 days (c) After 1500 days (d) After 1200 days 8

**207.** men and 1 woman can complete a piece of work in 14 days, while 4 women and 2 men can do the same work in 8 days. If a man gets 90 per day, what should be the wages per day Of a woman? (d) 48 72 e 135 [CDS (I) Paper 10 Q30]

(a) HCF

(b) HCF

(c) HCF

(d) None Of the above If b is the largest square divisor Of c and a2 divides C, then which one of the following is correct?

**208.** A and B are two taps which can fill a tank individually in 10 minutes and 20 minutes respectively. However, there is .a leakage at the bottom, which can empty a filled tank in 40 minutes. If the tank is empty initially, how much time *c will both the taps take to fill the tank with leakage ? [CDS (I) Paper 14 Q17]

(a) 2 minutes (b) mihutes (c) 5 minutes (d) 8 minutes

**209.** By increasing the speed of his car by 15 km/hour, a person covers 300 km distance by taking an hour less than before. The original speed of the car [CDS (I) Paper 14 Q28]

(a) 45 km/hour (b) 50 km/hour (c) 60 kni/hour (d) 75 km/hour

**210.** When the winner wins the race, then the distance between the athlete in the second position to the athlete in the third mysition is [CDS (I) Paper 14 Q11]

(a) 1000m (b) 800m (c) 750m (d) 600 m

**211.** An aeroplane flying at a height of 3000 m passes vertically above another aeroplane at an instant when the angles of elevation of the two planes from some point on the ground are 600 and 450 respectively. Then the vertical distance between the two planes is [CDS (I) Paper 14 Q43]

(a) 100066-1 (b) 10000m

(c) 100+-6 (d) 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

**212.** In a race of 100 m, A beats B by 4 m and A beats C by 2 m. By how many metres (approximately) would C beat B in another 100 m race assuming C and B run with their respective speeds as in the earlier race ? (b) 204 (c) 208 (d) 3-2 16 [CDS (I) Paper 14 Q48]

(a) 100066-1 (b) 10000m

(c) 100+-6 (d) 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

**213.** A boy is cycling such that the wheels of the Cycle are making 140 revolutions per minute. If the radius of the wheel is 30 cm, the speed of the cycle is [CDS (I) Paper 14 Q55]

(a) 15-5 km/hour (b) 15*84 kn/hour (c) 16 km/hour (d) 16-36 km/hour

**214.** A river 2-5m deep and 45m wide is flowing at the speed of km/hour. The amount of water that runs into the sea per minute is [CDS (I) Paper 14 Q80]

(a) 6650 (b) 6750 (c) 6850 (d) 6950 - A 28

**215.** A man walking at 5 km/hour noticed that a 225 m long train coming in the opposite direction crossed him in 9 seconds. The speed Of the train is [CDS (I) Paper 15 Q22]

(a) 75 km/hour (b) 80 km / hour (c) 85 km / hour (d) 90 kin/ hour w/ 54A

**216.** 25. sin 350 cos 550 2 COS 550 t 2 sin30a is equal sin 350 A cyclist moves non-stop from A to B, a distance of 14 km, at a certain average speed. If his average speed reduces by I km per hour, he takes 20 minutes more to cover the same distance. The original average speed Of the cyclist is [CDS (I) Paper 15 Q23]

(a) I only (b) 2 only

(c) Both I and 2 (d) Neither I nor 2 710 - 510 is divisible by 10

**217.** A boy saves Rs. 4*65 daily. What is the least number of days in which he will be able to save an exact number of rupees ? [CDS (I) Paper 16 Q11]

(a) 10 (b) 20 (c) 21 (d) 25

**218.** A can do 50% more work than B in the same time. B alone can do a piece of work in 30 hours. B starts working and had already worked for 12 hours when A joins him. How many hours should B and A work together to complete the remaining work ? [CDS (I) Paper 16 Q17]

(a) 6 hours (b) 12 hours (c) hours (d) 7-2 hours

**219.** When the speed 0fa train is increased by 20%, it takes 20 minutes less to cover the same distance. What is the time taken to cover the same distance with the original speed ? [CDS (I) Paper 16 Q18]

(a) 140 minutes (b) 120 minutes (c) 100 minutes (d) 80 minutes

**220.** A person can row downstream 20 km in 2 hours and upstream 4 km in 2 hours. What is the speed of the current ? [CDS (I) Paper 16 Q26]

(a) 2 km/hour (b) 2-5 km,'hour (c) 3 km/hOur (d) 4 knvhour

**221.** A tank can be filled by pipe X in 2 hours and pipe Y in 6 hours. At IO a.m. pipe X was opened. At what time will the tank be filled if pipe Y is opened at 11 a.m. ? [CDS (I) Paper 16 Q37]

(a) 12:45 hours (b) 5:00 p.m. (c) 11:45 a.m. (d) 11:50 a.m.

*— SPACE FOR ROUGH WORK —*

**222.** Ihe wheels Of a car are of diameter 80 cm each. The car is travelling at a speed of 66 km,'hour. What is the number of complete revolutions each wheel makes in IO minutes ? [CDS (I) Paper 16 Q67]

(a) 4275 (b) 4350 (c) 4375 (d) 4450

**223.** A motorist travels to a place 150 km away at an average speed of 50 km/hour and returns at 30 km/hour. What is the average speed for the whole journey ? [CDS (I) Paper 16 Q99]

(a) 35 km,fhour (b) 37 km,fhour (c) 37-5 (d) 40 km/hour

**224.** A of gmiM is *uÖ 24 mn 48 days. How long will wne stock lut 160 ma ? [CDS (I) Paper 21 Q25]

(a) 17 12 32 19 21 24. N is by 17, is equal to 182. ne b«wen the quotiai md is 175. is åe value of N ?

(b) 2975

(c) 3094

(d) 3101

**225.** A has avaage of 510 on md 240 days. What is the mnnb« Of visitors in a of 30 days b%inning Snu*y ? [CDS (I) Paper 21 Q81]

(a) 276 (b) 282 (c) 285 (d) 375 C -

**226.** A train takes two hours less for a journey of 300 km if its speed is increased by 5 knvhr from its usual speed. What is its usual speed ? [CDS (I) Paper 22 Q29]

(a) 50 knvhr (b) 40 km/hr (c) 35 knvhr (d) 25 km/hr

**227.** Jails from which more than half of the trained convicts are offered jobs, are [CDS (I) Paper 22 Q40]

(a) A, B and C (b) A, B and D

(c) A, E and F 41. The number of three digit numbers (d) Six

**228.** In covering certain distance, the average speeds of X and Y are in the ratio 4 : [CDS (I) Paper 22 Q44]

(a) A, B and C (b) A, B and D

(c) A, E and F 41. The number of three digit numbers (d) Six

**229.** Out of 4 identical balls of radius r, 3 balls are placed on a plane such that each ball touches the other two balls. The 4th ball is placed on them such that this ball touches all the three balls. What is the distance Of centre Of 4th ball from the plane ? [CDS (I) Paper 22 Q87]

(a) 7-5 m 86. What is the area of the largest square plate cut from a circular disk Of radius one unit ?

(b) 4 square units

(c) 245 square units

(d) r square units

**230.** men are supposed to complete a work in 10 days. Aner working for 5 days, they realise that only one-fourth of the work is done. How many more men they need to employ to finish the work on time ? [CDS (I) Paper 25 Q6]

(a) 40 (b) 30 (c) 20 (d) 15

**231.** Let work done by (3n - 1) men in (2n + 1) days be x and work done by (3n + 1) men in (4n - 3) days be y. If x : y = 6: 11, then what is the value of [CDS (I) Paper 25 Q28]

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2 only when Ware correct ? 28. Let work done by

**232.** Sheela can stitch a suit in 2 days. while Meena can stitch a suit in (hys. How many days will both take in stitching 30 suits ? [CDS (I) Paper 25 Q39]

(a) 32 days (b) 33 days (c) 35 days (d) 40 days

**233.** At the end of the time he received 11,400. How many days did he work ? [CDS (I) Paper 25 Q100]

(a) 20 (b) 21 (c) 24 (d) 25

**234.** A person rode one-third of a journey at 60 km/hr, one-third at 50 km/hr and the rest at 40 km/hr. Had the person ridden half of the journey at krnfhr and the rest at 40 km/hr, he would have taken 4 minutes longer to complete the journey. What distance did the person ride ? [CDS (I) Paper 25 Q44]

(a) 180 km (b) 210 km (c) 240 km (d) 300 km

**235.** Two circles touch externally. The sum of their areas is 411t square cm. If the distance between their centres is 9 cm, then what is difference between their diameters ? (b) I *5 cm (c) 2 cm [CDS (I) Paper 25 Q62]

(a) sin 9

(b) cose

(c) tan 0

(d) cot O 59. If A, B and C are interior angles of a triangle ABC, then what is equal to ? 2 A+B+C

**236.** A car travels from A to B at a speed Of 40 km/hr, travels back from B to A at a speed of 30 km/hr and again goes from A to B at a speed of 60 km/hr. What is the average speed of the car? [CDS (I) Paper 26 Q19]

(a) km/hr (b) 42 km/hr (c) 40 km/hr (d) km/hr

*— SPACE FOR ROUGH WORK —*

**237.** An following? aeroplane is Observed to be approaching the airport. It is at a distance Of IO km from the point of observation on the ground and makes an angle Of elevation e. If the aeroplane is at a height Of 8 km above the ground, then Which one of the following is corre ct? [CDS (I) Paper 26 Q53]

(a) 120'

(b) 135'

(c) 150* 52. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of SS. What is

(d) following? aeroplane is Observed to be 56. If tane + sece 3, then What is the value of 3 tan8+ 9sece?

**238.** The average of the temperatures recorded at noontime from Monday to Sunday is 31 oc. Ifthe lowest tempera- ture recorded is 300C, then what is the maximum of temperature that is possible to record at noontime on any one of the days [CDS (I) Paper 31 Q77]

(a) 340C (b) ,350C (c) 360C (d) 370C 33

**239.** X can do a piece of work in 4 days and Y can do the same piece of work in 6 days. Which of the following statements is/ are correct ? I. If X and Y work alternately starting with X, then the piece of work will be finished on the fifth day. [CDS (I) Paper 34 Q14]

(a) 3 months

(b) 6 months

(c) 9 months

(d) 1 year 13. If x varies directly as y, then which of the following is/ are correct ? I.

**240.** If X and Y work alternately starting with Y, then the piece of work will be finished in less than 5 days. Select the answer using the code given below : [CDS (I) Paper 34 Q11]

(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

**241.** men working 8 hours a day can completely build a wall of length 100 m, breadth 20 cm and height 5 m in 10 days. How many days will 16 men working 10 hours a day require to build a wall of length 200 m, breadth 60 cm and height 5 m ? [CDS (I) Paper 34 Q15]

(a) 18 (b) 24 (c) 27 (d) 36

**242.** Without stoppages, a train travels at an average speed of 60 km/h and with stoppages it covers the same distance at an average speed Of 48 km/h. HOW many minutes per hour does the train stop ? [CDS (I) Paper 34 Q17]

(a) 10 minutes (b) 12 minutes (c) 1 S minutes (d) 20 minutes

**243.** What is the greatest possible speed at which a person can walk 13-3 km and 209 km so that the time (in hours) required in each case is a whole number ? [CDS (I) Paper 34 Q26]

(a) 1-7 km/h (b) 1-9 km/h (c) km/h (d) 2*3 km/h

**244.** A locomotive engine can go 40 km/hr. Its speed gets reduced by a quantity that varies directly as the square root of the number of wagons attached. It is known that its speed becomes 16 km/hr if 36 wagons are attached. What is the smallest number of wagons with which the engine is unable to move ? [CDS (I) Paper 34 Q34]

(a) 100 (b) 81 (c) 64 (d) 49

## SECTION : ALGEBRA, POLYNOMIALS & LINEAR EQUATIONS (193 QUESTIONS)

**245.** From a rectangular metal sheet Of sides 25 cm and 20 em, a circular sheet as large as possible is cut-off. What is the area of the remaining sheet ? [CDS (I) Paper 1 Q49]

(a) 186 (b) 144 (c) 93 cm2 (d) 72 cm2

**246.** (d) x - y = 10 [CDS (I) Paper 1 Q76]

(a) x - y = 10 78. 79. 80. 81. 82. 83. 84. sin x cos x 1/2 t, sin x - cos x DE t

(b) *rut A log IO

(c) Tt/2

(d) Tt/3

**247.** Assertion [CDS (I) Paper 1 Q7]

(a) Both A and R are individually true and R is the correct explanation of A

(b) Both A and R are individually true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true 98. Assertion

**248.** What was the original wage bill of the factory if the present bill is Rs. 5,000/- ? (b) Rs. 6000/- (c) Rs. 8,000,'- (d) None of the above [CDS (I) Paper 3 Q2]

(a) 4 days (b) 5 days (c) 6 days (d) days 33. If of

**249.** If a quantity y varies as the sum of three quantities of which the first varies as x, the second varies as - x+x2, the third varies as x3 - p then what is y equal to ? [CDS (I) Paper 3 Q35]

(a) where k is a constant

(b) kx+lx2+nu3, where k, l, m are constants

(c) kP where k is a constant

(d) kx where k is a constant - A 10

*— SPACE FOR ROUGH WORK —*

**250.** What is the „ + sin275ø ? + sin2250 + [CDS (I) Paper 3 Q41]
(a) tan2150 + + tan2250 tan2750
(b) cos2150 + cos220e + cos2250 . cos2750
(c) + cot225* - + cot2750
(d) sec2200 + sec2250

**251.** A semi circular thin sheet of a of diameter 28 cm is bent and an open conical cup is made. What is the capacity of the cup ? 1000 [CDS (I) Paper 3 Q76]
(a) cm3
(b) 3006 cm3
(c) cm3 1078
(d) cm3 - A 22

**252.** Smaller lead shots are to be prepared by using the material of a spherical lead shot of radius I cm. Some possibilities are listed in the statements given below: . I. The material is just sufficient to prepare 8 shots each of radius cm. [CDS (I) Paper 3 Q77]
(a) 4-5 74. What is volume of the frustum of a cone with height h and radii r, , ?
(b) 9th
(c) 15m
(d) I *5m

**253.** A shot of radius 0-75 cm and a second shot of radius 0-8 cm can be prepared from the available materiaL Which of the above statements is/ are correct ? [CDS (I) Paper 3 Q2]
(a) 1 Only
(b) 2 Only
(c) Both I and 2
(d) Neither nor 2

**254.** The HCFof x - y and xlO - y10 is Which of the statements given above Correct ? are [CDS (I) Paper 3 Q4]
(a) 1 and 2
(b) 2 and 3
(c) 1 and 4
(d) 2 and 4 26

**255.** What is the sum of the squarés of all the six numbers ? [CDS (I) Paper 3 Q98]
(a) 40
(b) 42
(c) 45
(d) 48

**256.** What are the factors Of [CDS (I) Paper 4 Q4]
(a) (x - 2y - 4) and (x - 2Y+2)
(b) (x - y and (x - 4 y + 4)
(c) (x - y and (x - 4 y - 4)
(d) (x (x +2Y+2) 26k /

**257.** Which one Of the following is a non-termmating and repeating decimal ? [CDS (I) Paper 4 Q30]
(a) 13/8
(b) 3/16
(c) 3/11
(d) 137/25

**258.** ABC BC D ZACD- 1200, ZABC- -2? [CDS (I) Paper 4 Q51]
(a) 47. 48. 49. 14 sm I - sine
(b) 9 - tan 9
(c) sece+ tan 8
(d) cosece + cot 9

**259.** What is the average marks Of all the candidates? [CDS (I) Paper 4 Q77]
(a) 758
(b) 75-8
(c) 760.
(d) 768'.

**260.** If two tridngles have equal area, then they are siinilar. Which Of the statements given above is/are corect? [CDS (I) Paper 4 Q2]
(a) I only
(b) 2 only I
(c) Both I and
(d) Neither I nér 2

**261.** AD2 - 3CD2 AC2 = AD2 [CDS (I) Paper 5 Q85]
(a) 121-7 121-75
(b) 121-75 ABC t, X Y AX.AYur
(c) 16 2
(d) 20 cm 2

**262.** What is the sum of their reciprocals ? [CDS (I) Paper 6 Q20]
(a) 2
(b) 2
(c) 2
(d) 2-562 + b2

**263.** In the figure given above, AB is parallel to CD. If Z DCE = x and Z ABE = y, then what is Z CEB equal to ? (c) x + y - (11/2) 980 1440 Iri the figure given above, AB is parallel to CD. If Z BAF 980 and AFC = 1440, then what is L ECD equal to ? [CDS (I) Paper 6 Q95]
(a) 620
(b) 640
(c) 820
(d) 840

**264.** How many degrees are there in an angle which equals two-third of its complement ? [CDS (I) Paper 6 Q96]
(a) 360 450
(b) 48
(c) 600 97. 470 670 N In the figure given above, PQ is parallel to RS. What is Z NMS equal to ?
(d) 270

**265.** A number consists of two digits. The sum of the digits is IO. On reversing the digits of the number, the number decreases by [CDS (I) Paper 7 Q4]
(a) 21
(b) 24
(c) 36
(d) 42 Which one among the following is not correct ?

**266.** Colour of the eye Of the person [CDS (I) Paper 7 Q2]
(a) 21
(b) 24
(c) 36
(d) 42 Which one among the following is not correct ?

*— SPACE FOR ROUGH WORK —*

**267.** Number of times a }xrson correctly hits a target in ten attempts Which of the following in respect of the above is correct ? [CDS (I) Paper 7 Q3]
(a) I is a continuous variable, 2 is not a variable and 3 is a discrete variable
(b) 1 is a continuous variable, 2 and 3 are discrete variables
(c) I and 3 are discrete variables and 2 is not a variable
(d) *l, 2 and 3 all are discrete variables Consider the followiög *statements :

**268.** The data collected by investigator to be used by himself are called primary data [CDS (I) Paper 7 Q1]
(a) 21 (b) 24
(c) 36 (d) 42 Which one among the following is not correct ?

**269.** Consider the following : l. 10+ I [CDS (I) Paper 7 Q66]
(a) p (b) 3 and - 4 (c) - 3 and - 4 (d) - 3 and 4

**270.** If a set Acontains 60 elements and another set B contains 70 elements and there are 50 elements in common, then how many elements does A u B contain ? [CDS (I) Paper 7 Q100]
(a) 130 (b) 100 (c) 80 (d) 70 28

**271.** If two factors of - 2a3 - 9a2 + 2a + 8 are (a + l) and (a - l), then what are the Other two factors ? [CDS (I) Paper 8 Q24]
(a) (a - 2) and (a + 4) (b) (a + 2) and (a + 4) (c) (a + 2) and (a - 4) (d) (a - 2) and (a - 4)

**272.** If 0.764 y 1.236 x, then what is the Y¯x value Of [CDS (I) Paper 8 Q26]
(a) x- 10 (b) x = 11
(c) x = 11.5 If 0.764 y = 1.236 x, Y - x value Of (d) 0.764

**273.** Which of the following is/are correctly matched ? [CDS (I) Paper 8 Q99]
(a) Mean Median (b) Mean Mode
(c) Mean > Median (d) Mean > Mode What is the harmonic mean 25, 40 and 50 ?

**274.** Educational qualification of the person Continuous variable Attribute 28 Select the correct answer using the code given below * [CDS (I) Paper 8 Q2]
(a) I only (b) 2 only (c) Both I and 2 (d) Neither 1 nor 2

**275.** The Of two number' is 6 10 12 15 Of N:uation* y O a unique x unique X infinite solutions -2+4 Za-l 2.3 - 28-1 2.8 - '28 + 292 *a + - Qy) only O.OETA [CDS (I) Paper 9 Q25]
(a) k = 2 (b) k = 4 (c) k = -2 (d) k = ±4

**276.** 3bc c c Sbe + b [d) 3be 4 b3 + then what is x What ig the value Of O ne Ot factors tf - 5x2 - 81 ia ta) equgi to ? a unique x a unique solution x - infinite solutions 2.3-28-1 2.8-'28 291 + [CDS (I) Paper 9 Q18]
(a) x = 1, y = 2 (b) x = 2, y = 1 (c) x = 3, y = 1 (d) x = 1, y = 3

**277.** Which the Only 2 only Neither Q-DETA.N.FMN Stet-em correct ? [CDS (I) Paper 9 Q1]
(a) x = 1, y = 2 (b) x = 2, y = 1 (c) x = 3, y = 1 (d) x = 1, y = 3

**278.** Which of the above statements is/are correct? [CDS (I) Paper 10 Q11]
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither nor 2 Which one is one of the factors of (a) (c) / 48A

**279.** If then what*is one of the values of x equal to? [CDS (I) Paper 10 Q13]
(a) 15 (b) 25
(c) 20 (d) 30 9. 10. 11. What is equal to?

**280.** The quantity which must be added to (1 - x 2) to obtain x 3 is 2X3 + 3X2 + I 2x3+x + x - 1 (b) Consider the following statements * [CDS (I) Paper 10 Q14]
(a) 15 (b) 25
(c) 20 (d) 30 9. 10. 11. What is equal to?

**281.** If x Y = 4 and + _ = 1 then what is x + y equal to? [CDS (I) Paper 10 Q20]
(a) O only (b) 20 only (c) both 10 and 20 (d) neither IO nor 20 If

**282.** How many 200 mm lengths can be cut from IO m of ribbon? [CDS (I) Paper 10 Q28]
(a) 50 (b) 40 (c) 30 (d) 20

**283.** What is 262 +972 equal to? @ 272 +932 (b) 342 +932 (C) 822 +412 (d) 792 + 622 [CDS (I) Paper 10 Q29]
(a) HCF
(b) HCF
(c) HCF
(d) None Of the above If b is the largest square divisor Of c and a2 divides C, then which one of the following is correct?

*— SPACE FOR ROUGH WORK —*

**284.** If a equation xa - x - 1 O, then what is equal to? [CDS (I) Paper 10 Q45]
(a) 56 (b) 60 cm (c) 64 cm (d) 1 only

**285.** ADM DCM ABCD t, Afa AB-CDtl [CDS (I) Paper 12 Q1]
(a) cos x (b) sin X + cos x
(c) sin x - cos x 63. * l. sin r > sin 1 2. cos l' cos I (d) 4q 66. 67.

**286.** If a Star figure is formed by elongating the sides of a regular pentagon, then the measure Of each angle at the angular points of the star figure is [CDS (I) Paper 13 Q56]
(a) 360 (b) 350 (c) 320 (d) 300

**287.** A tin of oil was - full. When 6 bottles of oil were taken out from this tin and 4 bottles of oil were poured into it, it was - fill]. Oil of. how many EX)ttles can the tin contain ? (All bottles are of equal volurne) [CDS (I) Paper 14 Q11]
(a) 35 (b) 40 (c) 45 (d) 50

**288.** Consider the following in respect of the equation y = l. if x > 1 2. if x < 1 3. y exists for all values of x Which of the above statements is/are correct ? [CDS (I) Paper 14 Q13]
(a) I only (b) 2 only
(c) I and 2 only l, 2 and Y is/are 14. If (d) 2 minutes

**289.** If k=x - y+2z where - 2<-r$l, - l y s 2 and 3 S z «*then which one of the following is correct ? (b) 5<k$11 [CDS (I) Paper 14 Q20]
(a) 43-2%
(b) 56-8% 20. If k=x - y+2z where - 2<-r$l, - l y s 2 and 3 S z «*then which one of the following is correct ?
(c) 5<k$11 21. The number of pairs
(d) Zero

**290.** In a race A, B and C take part. A beats B by 30 m, B beats C by 20 m and A beats C by 48 m. Which of the following is / are conect ? *l. The length Of the race is 300 m. [CDS (I) Paper 14 Q24]
(a) 43-2%
(b) 56-8% 20. If k=x - y+2z where - 2<-r$l, - l y s 2 and 3 S z «*then which one of the following is correct ?
(c) 5<k$11 21. The number of pairs
(d) Zero

**291.** Select the correct *answer using the code given below * [CDS (I) Paper 14 Q42]
(a) I only (b) 2 only (c) Both I and 2 (d) Neither I nor 2 8

**292.** If and b. c 2a : 3b : 7c is equal to [CDS (I) Paper 14 Q30]
(a) 42 : 105 : 320 (b) 15 : 21 : 35
(c) 6: 15:40 (d) 30; 21 : 350* 10 = 7: 8, then

**293.** If O is an acute angle and sinecos0= 2cos30 - 1-5cosO, then what is sine equal to ? 6-1 4 1-6 (b) 4 4 4 [CDS (I) Paper 14 Q40]
(a) tan29 (b) sin2e COS 9 38. is equal to I - sine tang - I tang +1 1 + sine
(c) coso tano + I tang - I (d) sin 6 39. If tan

**294.** Consider the following statements : 1. sin660 is less than cos660. 2„ sin260 is less than cos260. Which of the above statements is/are corect ? [CDS (I) Paper 14 Q41]
(a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither I nor 2

**295.** If a and b are positive, then the relation sino 3b (b) Byssible only if a b (c) possible if a > b (d) possible if a < b * 4 [CDS (I) Paper 14 Q42]
(a) tan29 (b) sin2e COS 9 38. is equal to I - sine tang - I tang +1 1 + sine
(c) coso tano + I tang - I (d) sin 6 39. If tan

**296.** A pole is standing erect on the ground which is horizontal tip of the_pole is tied tight with a roly of length VI 2 m to a point on the ground. If the rope is making 300 with the horizontal, then the height of the pole is [CDS (I) Paper 14 Q44]
(a) 100066-1 (b) 10000m
(c) 100+-6 (d) 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

**297.** If tano + = 2, then tang is equal to 3 4 5 4 2 5 (d) 2 - A 16 [CDS (I) Paper 14 Q45]
(a) 100066-1 (b) 10000m
(c) 100+-6 (d) 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

**298.** If the digits are reversed,' the number is increased by [CDS (I) Paper 14 Q7]
(a) 100066-1 (b) 10000m
(c) 100+-6 (d) 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

**299.** A hollow cylindrical drum has internal diameter of 30 cm and a height of 1 m. What is the maximum number of cylindrical boxes of diameter 10 cm and height IO cm each that can be packed in the drum ? [CDS (I) Paper 14 Q52]
(a) 60 (b) 70 (c) 80 (d) 90

*— SPACE FOR ROUGH WORK —*

**300.** The petpendiculars drawn from any point on OM to AB and AC are always equal Which of the above statements are correct ? [CDS (I) Paper 14 Q3]

(a) - 1 and 2 only (b) 2 and 3 only (c) 1 and 3 only (d) l, 2 and 3

**301.** D In the above figure, ABCD is a parallelo- gram. p is a point on BC such that PB : PC=I : [CDS (I) Paper 14 Q83]

(a) 20 square t

(b) 30 square unit

(c) 40 square unit

(d) None of the above 84. Chord CD intersects the diameter AB of a circle at riW1t angle at a m

**302.** The value of (063+0-37) is 100 99 1000 (d) 999 30 In the above figure, ABCD is a parallelo- gram. p is a point on BC such that PB : PC=I : [CDS (I) Paper 14 Q87]

(a) 20 square t

(b) 30 square unit

(c) 40 square unit

(d) None of the above 84. Chord CD intersects the diameter AB of a circle at riW1t angle at a m

**303.** The difference of maximum values of the expressions (6 + 5x - x2) and (y - 6 - y2) for any real values of x and [CDS (I) Paper 14 Q90]

(a) 16 (b) 17 (c) 18 (d) 19

**304.** A clock is started at noon. By 10 minutes past 5, through what angle, the hour hand rnoves ? [CDS (I) Paper 14 Q91]

(a) 1600 (b) 1450

(c) 1500 (d) 1550 - A 32 92. minimum value of cos2x + cos2y - cos2z

**305.** The average weight of students in a class is 43 kg. Four new students are admitted to the class whose weights are 42 kg. 36*5 kg, 39 kg and 42-5 kg respectively. Now the average weight, of the students of the class is 426 kg. The number of students in the beginning was [CDS (I) Paper 14 Q96]

(a) 24 (b) a - 25 (c) 28 (d) a - 1 25

**306.** Four years ago, the average age of A and B was 18 years. Now the average age Of A, B and C is 24 years. After 8 years, the age of C will be [CDS (I) Paper 14 Q97]

(a) 32 years (b) 28 years

(c) 36 years (d) 40 years - A (a) (b) (c) 34

**307.** The solution Of the inequation is ( given that x O) -I-JS -ltvs (c) [CDS (I) Paper 15 Q37]

(a) 38. Which of the points P (b) Q and R o nly (c) P and R o nly (d) P and Q o Illy

**308.** Which of the points P(5, - I), QI 3, - 2) and R(I, I) lie in the solution of the system Of inequations x + y 4 and x-y>2? [CDS (I) Paper 15 Q38]

(a) Q and R o nly (b) P and R o nly (c) P and Q o Illy (d) P, Q and R

**309.** There are three brothers. The sums of ages of two Of them at a time are 4 years, 6 years and 8 years. The age difference between the eldest and the youngest is [CDS (I) Paper 15 Q42]

(a) 3 years (b) 4 years (c) 5 years (d) 6 years

**310.** If p = cotB+ tane and q = sece - then (p2q)5 - (q2p)S is equal to [CDS (I) Paper 15 Q64]

(a) 2 -Be , thenC+2B

(b) 0-1 0+1 cos26 - 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e

(c) only

(d) 2 only

**311.** If (c) 0-1 0+1 cos26 - 3cos8+2 sin2 where O < e < then which of the following statements is/ are correct? I. There are two values of e satisfy ing the above equation. [CDS (I) Paper 15 Q66]

(a) 2 -Be , thenC+2B

(b) 0-1 0+1 cos26 - 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e

(c) only

(d) 2 only

**312.** 260' is satisfied by the above Select the correct answer using the code given below. (b) (d) only 2 only Both I and 2 Neither I nor 2 / S4A [CDS (I) Paper 15 Q2]

(a) 2 -Be , thenC+2B

(b) 0-1 0+1 cos26 - 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e

(c) only

(d) 2 only

*— SPACE FOR ROUGH WORK —*

**313.** There are five lines in a no two of which are parallel. The maximum number of points in which they can intersect is (c) 4 10 None Of the above [CDS (I) Paper 15 Q69]

(a) 2 -Be , thenC+2B

(b) 0-1 0+1 cos26 - 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e

(c) only

(d) 2 only

**314.** The value of the expressi n (243 64712 243x243+647x 7 is equal to [CDS (I) Paper 15 Q98]

(a) AM < Median (b) GM Median

(c) AM = Me dian Suppose for Which one of the corre ct? (d) GM>1/6

**315.** Ifa < b, then - Select th e correct answer sing the code given below. (b) 2 only (c) 3 only (d) 2 and 3 [CDS (I) Paper 15 Q2]

(a) AM < Median (b) GM Median

(c) AM = Me dian Suppose for Which one of the corre ct? (d) GM>1/6

**316.** If e - y + z, = z + x and then what is the value of - A [CDS (I) Paper 16 Q4]

(a) x = 100 (b) x = IO (c) 330 (d) 340

**317.** What would be the maximum value of Q in the equation 5P9+ 3R7 + 2Q8= 1114? [CDS (I) Paper 16 Q5]

(a) x = 100 (b) x = IO (c) 330 (d) 340

**318.** In an examination, a student was asked to divide a certain number by [CDS (I) Paper 16 Q7]

(a) x = 100 (b) x = IO (c) 330 (d) 340

**319.** By mistake he multiplied it by 8 and got the answer 2016 more than the correct answer. What was, the number ? [CDS (I) Paper 16 Q8]

(a) 252 (b) 256 (c) 258 (d) 260

**320.** What is to ? 62 +72 +8 7 + 46 - 2 + 92 + 4 + 26 = z + x and then what is the value of - A [CDS (I) Paper 16 Q3]

(a) x = 100 (b) x = IO (c) 330 (d) 340

**321.** Let A = {7, 8, 9, 10, 11, 12} and B = {7, IO, 14, 15}. What is the number of elements in (A - B) and (B - A) respectively ? [CDS (I) Paper 16 Q10]

(a) 2 and 4 (b) 4 and 2 (c) 2 and 2 (d) 4 and 4

**322.** If a and are the two zeros of the polynomial 25x2 - 15x + 2, then what is a quadratic polynomial whose zeros are (2a)-l and (2ß)-l ? [CDS (I) Paper 16 Q14]

(a) x2+30x+2 (b) 8x2-30X+25 (c) 8x2-30x (d) x2+30x

**323.** Both the numbers may be com- posite. Select the correct answer using the code given below : [CDS (I) Paper 16 Q3]

(a) 1 and 2 only (b) 2 and 3 Only (c) 1 and 3 only (d) l, 2 and 3

**324.** The number Of digits in 330 is n and it is given that log103 -04771. What is the value Of n ? [CDS (I) Paper 16 Q32]

(a) 13 (b) 14 (c) 15 (d) 16 - A 10

**325.** Into a conical tent Of radius 8*4m and vertical height m, how many full bags of wheat can be emptied, if space required for the wheat in each bag is [CDS (I) Paper 16 Q45]

(a) 264 (b) 201 (c) 132 (d) 105

**326.** What is the number of rounds that a wheel of diameter -N m will make in traversing 7 km ? [CDS (I) Paper 16 Q49]

(a) 3300 (b) 3500 (c) 4400 (d) 4900

**327.** Rain water from ao roof 22 m x 20 m drains into a cylindrical vessel having diameter of base 2 m and height 3-5 m. If the vessel is just full, what is the rainfall ? [CDS (I) Paper 16 Q57]

(a) Rs. 8

(b) Rs. 9

(c) Rs. 13

(d) Rs. 14 53. If a quadrilateral has an Inscribed then the sum of a pair Of opposite sides equals

**328.** If sin6+cos9= 2 sing - coso equal to ? [CDS (I) Paper 16 Q71]

(a) 4275

(b) 4350

(c) 4375

(d) 4450 68. What is the area of a triangle with sides Of length 12 cm, 13 cm and 5 cm ?

**329.** A cricketer has a certain average Of IO innings. In the eleventh inning he scored 108 runs, thereby increasing his average by 6 runs. What is his new average ? [CDS (I) Paper 16 Q89]

(a) 42 (b) 47 (c) 48 (d) 60

**330.** Number of credit cards held by an Indlvldual can be treated as [CDS (I) Paper 16 Q91]

(a) Qualitative data (b) Discrete data (c) Categorical data (d) None of the above

*— SPACE FOR ROUGH WORK —*

**331.** The classes of type 15-20, 20-25, 25 - 30 etc. are inclusive classes. Which Of the above statements is/are correct ? [CDS (I) Paper 16 Q2]
(a) I only (b) 2 only (c) Both I and 2 (d) Neither I nor 2

**332.** If D is the number Of degrees and R is the number Of radians in an angle O, then which one Of the following is correct ? (b) (c) (d) = IBOR 1801) c 100 1 *50 200 50 20 What is the minimum value Of 9 tan2 + 4 cot2 9 ? It is known that the average price per pencil is 1*25. M'llat is the value Of x ? [CDS (I) Paper 17 Q60]
(a) 10 (b) e 30 (c) e 40 (d) 60 (a) (c) (d) 6 9 12 13

**333.** For value of k x3 + - + 6 be C [CDS (I) Paper 21 Q16]
(a) be 2 one Of the 61%wing ? 13. HCF numbes is 12. Wych me Of xver LCM ?
(b) 80
(c) 60 36 *
(d) 24 14. WEch Of

**334.** WEch Of [CDS (I) Paper 21 Q14]
(a) 1,2and3 (b) I 2 Mily
(c) 2 3 mxly (d) I 3 Mly IS. TIE rüe of two is it is But in Of the half is in dre

**335.** N is by 17, is equal to 182. ne b«wen the quotiai md is 175. is åe value of N ? [CDS (I) Paper 21 Q24]
(a) 2975 (b) 3094 (c) 3101 (d) 3269

**336.** Which orz of åe mllowing is ca*ræt in Of a light riangle ? [CDS (I) Paper 21 Q29]
(a) Its lia ixuide O
(b) Its lies on the triülgle
(c) It 30. Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ?
(d) AB<RX

**337.** Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ? [CDS (I) Paper 21 Q30]
(a) Its lia ixuide O
(b) Its lies on the triülgle
(c) It 30. Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ?
(d) AB<RX

**338.** If = l, then What is value of + 3iniOx + + ? *13 [CDS (I) Paper 21 Q32]
(a) Its lia ixuide O
(b) Its lies on the triülgle
(c) It 30. Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ?
(d) AB<RX

**339.** If Of fre lalgths of is 12 what is arn of ? [CDS (I) Paper 21 Q2]
(a) 9 an-2 (b) 12 cm2 (c) 16 (d) 25 cma C -

**340.** If one Of a aingle (with all sides intega*s) is 15 an, thm maximum Of uiangle ? [CDS (I) Paper 21 Q63]
(a) 240 cm (b) 225 (c) 113 cm (d) 112 cm

**341.** If p is relatively to of g r, p is relatiüely prime to poduct qr. [CDS (I) Paper 21 Q1]
(a) 1 only (b) 96" cm2 (c) 2 only (d) Boå 1 2 69T cm2

**342.** Rmi Five yen back, the of Radha Was tima that of Rni, but buk age of WE pf Rmi. is age them' ? (c) 10 (d) 11 [CDS (I) Paper 21 Q73]
(a) 1 only (b) 96" cm2 (c) 2 only (d) Boå 1 2 69T cm2

**343.** Whatis åe qure root of 16+66 ? 0) 4-47 (c) 3+4i (d) 3-N5 [CDS (I) Paper 21 Q76]
(a) 3+4i
(b) 3-N5 77. What is the numb' in 125, 823 md 920 ? [Givn hg102 = 0-301, log. - &477, 10867 - 0845]
(c) 276
(d) 282

**344.** How pairs Of ze such that åe of thü Nuar - 35 ? [CDS (I) Paper 21 Q90]
(a) -1.2
(b) -1,-2 4 90. How pairs Of 15 ze such that åe of thü Nuar - 35 ? 91. If
(c) 21
(d) 19 value of x2+5x+ O 94. mtio of

*— SPACE FOR ROUGH WORK —*

**345.** Which one of the following is the largest number among 22222, 22222, 22222, 22222 ? [CDS (I) Paper 22 Q7]
(a) a prime number
(b) not a prime, but power of a prime
(c) not a power of a prime, but a composite even number
(d) not a of a prime, but a composite odd number 2. Two unequal pairs of numbers satisfy the following conditions :

**346.** If 10500 and y what is logy equal to ? [CDS (I) Paper 22 Q21]
(a) 4-2500 (b) 2-5625 (c) 1-5625 (d) 1-2500

**347.** If x varies asyz, theny varies inversely (x:) [CDS (I) Paper 22 Q24]
(a) 49x2 -f- 3x + 2 (b) 49x2 _ 3x + 2
(c) 49++2 (d) 59++2 24. If x varies asyz, theny varies inversely

**348.** What is the point on the xyplane satisfying 5x +2y 7xy and (b) (c) I Ox + 3y ? '6 6' 6' [CDS (I) Paper 22 Q26]
(a) 49x2 -f- 3x + 2 (b) 49x2 _ 3x + 2
(c) 49++2 (d) 59++2 24. If x varies asyz, theny varies inversely

**349.** The average height of students in the class is 4-6 feet. Ifthe average height of the boys in the class is feet, then what is the average height of the girls in the class ? [CDS (I) Paper 22 Q4]
(a) Less than 4-2 feet (b) More than 4*2 feet but less than 4-3 feet
(c) More than 4*3 feet but less than 4-4 feet (d) More than 44 feet but less than feet 11 A -

**350.** Let None of the above XYZ be a 3-digit number. Let S-XYZ+YZX+ZW. Which of the following statements Ware correct ? [CDS (I) Paper 22 Q43]
(a) A, B and C (b) A, B and D
(c) A, E and F 41. The number of three digit numbers (d) Six

**351.** If X takes 45 minutes more than Y to reach the destination, then what is the time taken by Y to reach the destination ? [CDS (I) Paper 22 Q5]
(a) 135 minutes (b) 150 minutes (c) 180 minutes (d) 225 minutes

**352.** A wheel makes 360 revolutions in one minute. What is the number of radians it tums in one second ? [CDS (I) Paper 22 Q51]
(a) 10 (b) 16i (c) 36 (d) 27

**353.** Let the (d) 17 A - [CDS (I) Paper 22 Q53]
(a) 10 (b) 16i (c) 36 (d) 27

**354.** HOW many cubic metre Of earth is to be dug out to dig a well of radius 14 m and [CDS (I) Paper 22 Q80]
(a) - AB (b) a AB2 (c) 1, 2,3 82. 83. 2 (d) 4

**355.** A ladder 5 m long is placed in a room so as to reach a point 4*8m high on a wall and on turning the ladder over to the opposite side of the wall without moving the base, it reaches a point 1 *4 m high. What is the breadth of the room ? (d) 7-5 m [CDS (I) Paper 22 Q85]
(a) 7-5 m 86. What is the area of the largest square plate cut from a circular disk Of radius one unit ?
(b) 4 square units
(c) 245 square units
(d) r square units

**356.** In the given figure, if 6 and = 5, then what is the value of x ? [CDS (I) Paper 22 Q93]
(a) 450 (b) 300 (c) 150 (d) 100

**357.** What is 8x equal to? 2 4 9 [CDS (I) Paper 24 Q17]
(a) Which Of the above statements is/are correct? (b) I only
(c) 2 only (d) Both I and 2

**358.** For what integral value of x is (b) [CDS (I) Paper 24 Q18]
(a) Which Of the above statements is/are correct? (b) I only
(c) 2 only (d) Both I and 2

**359.** 3 2 - + I = k 2 , then which One of the is a possible expression k P - 3X+I [CDS (I) Paper 24 Q19]
(a) Which Of the above statements is/are correct? (b) I only
(c) 2 only (d) Both I and 2

**360.** What is bc(a - b) (a - equal to ? [CDS (I) Paper 24 Q20]
(a) Which Of the above statements is/are correct? (b) I only
(c) 2 only (d) Both I and 2

**361.** + secB 8 - 6 tane [CDS (I) Paper 24 Q58]
(a) 11 (b) 22 (c) 77 (d) 99 k sece

*— SPACE FOR ROUGH WORK —*

**362.** Consider the -following data With regard to production Of cars (in lakhs) : Year 2015 Year 2016 Coun Coun Coun Coun 38 88 [CDS (I) Paper 24 Q91]

(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015?

(b) and E

(c) A, C and D only

(d) C D and E

**363.** The marks obtained by 5 students are 21, 27, 19, 26, [CDS (I) Paper 24 Q95]

(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015?

(b) and E

(c) A, C and D only

(d) C D and E

**364.** Which product is produced least over the years ? [CDS (I) Paper 24 Q97]

(a) Type 11 (b) Type 111 (c) Type (d) Type V ,'SOA

**365.** In Which one Of the following pairs of years, the difference in total number Of tablets produced between them is m inimum? [CDS (I) Paper 24 Q98]

(a) (2003, 2005) (b) (2001, 2005) (c) g003, 2004) (d) (2000, 2002)

**366.** What is the sum of the linear factors (in x and y) of the expression 2x-2 + - 3y2 ? [CDS (I) Paper 25 Q8]

(a) 2t-3y (b) 3x-2y (c) 3x + 2y (d) 2r+3y

**367.** For which values of k, does the equation x2 - kx + 2 0 have real and distinct solutions ? (b) only (c) k>2v5 only (d) k<-24i or k>2Vi [CDS (I) Paper 25 Q11]

(a) only (b) k>2v5 only

(c) k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is (d) I - aßy

**368.** If a + D + Y aß+ "y + ya, then what is (1 equal to? [CDS (I) Paper 25 Q12]

(a) only (b) k>2v5 only

(c) k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is (d) I - aßy

**369.** pencils, 6 notebooks and 7 erasers cost *250; whereas 6 pencils, 4 note- books and 2 erasers cost [CDS (I) Paper 25 Q16]

(a) only (b) k>2v5 only

(c) k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is (d) I - aßy

**370.** How many zeros are there in the product 150 x 249 x 348* x 501 ? [CDS (I) Paper 25 Q17]

(a) 262 (b) 261 (c) 246 (d) 235

**371.** If 5-t¯l then what is the value of x ? (b) logr02 (c) log105 (d) 210g105 [CDS (I) Paper 25 Q20]

(a) 30 (b) 45 (c) 60 (d) 75 23. What IS value of N?

**372.** If 96-64a3 3 = what is alt + 4a3 equal to ? then [CDS (I) Paper 25 Q21]

(a) 30 (b) 45 (c) 60 (d) 75 23. What IS value of N?

**373.** If A+B= equal to ? x2 - 4x+4 2x2 -7X+3 2x-1 2x2+7x-3 2x-1 then what is B 7 the 7 [CDS (I) Paper 25 Q22]

(a) 30 (b) 45 (c) 60 (d) 75 23. What IS value of N?

**374.** If then what is (a + b) : (a - b) equal to ? [CDS (I) Paper 25 Q24]

(a) 3 only (b) 5 only (c) - 3 or 3 (d) - 5 or 5

**375.** Consider the following inequalities : a2 - b2 a - b > - where a > b > O a2+b2 a a3+b3 a2+b2 only when 2. a2+b2 a+b Which of the above is/are correct ? [CDS (I) Paper 25 Q27]

(a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2

**376.** If ay - bx _ cx - az _ bz ¯ then which of the following is/are correct ? x+y+z_z 2. a+b+c¯c Select the correct answer using the code given below * [CDS (I) Paper 25 Q29]

(a) I only (b) 2 only (c) Both I and 2 (d) Neither 1 nor 2 9

**377.** Ihe present age of a father is equal to sum of the ages of his 4 children. Aner ten years the sum of the ages of the children will be 1-6 times the age of their father. What is the present age of father ? [CDS (I) Paper 25 Q34]

(a) 36 years (b) 40 years (c) 42 years (d) 45 years

**378.** Between 3 and 4 0'clock, both hour hand and minute hand will coincide past 3 0'clock between [CDS (I) Paper 25 Q38]

(a) 15-16 minutes (b) 16-17 minutes (c) 17-18 minutes (d) 18-19 minutes

**379.** A person saves more than he did the previous year. If he saves *2000 in the first year, in how many years will he save 170000 ? 13 [CDS (I) Paper 25 Q45]

(a) 16 years (b) 17 years (c) 18 years (d) 19 years

*— SPACE FOR ROUGH WORK —*

**380.** A hemispherical bowl of intemal radius 18 cm contains a liquid. The liquid is filled in small cylindrical bottles of internal radius 3 cm and internal height 4 cm. What is the number of bottles used to empty the bowl ? [CDS (I) Paper 25 Q67]

(a) 54 (b) 81 (c) 108 (d) 135

**381.** A hollow spherical shell is made of a metal of density 7 g/cm3. If its internal and extemal radii are 3 cm and 6 cm respectively, then what is the mass of the shell ? (take [CDS (I) Paper 25 Q68]

(a) 2772 g (b) 3322 g (c) 4433 g (d) 5544 g 19 19

**382.** What is AB2 + 4BQ2 equal to ? [CDS (I) Paper 25 Q86]

(a) 41*25cm2 (b) 42*25 cm2 (c) 43-75 cm2 (d) 44*25 cm2

**383.** What is AQ2 + CP2 equal to ? (b) (c) (d) 1-2 AC2 1-25 AC2 1-5 AC2 A - [CDS (I) Paper 25 Q87]

(a) 10*05 cm2 (b) 10*15cm2

(c) 11-05 cm2 (d) 11*15cm2 Consider the following for the next three

**384.** There are two categories of employees in 2019 whose total strength remained the same in 2020. What are these two categories ? [CDS (I) Paper 25 Q91]

(a) C and B (b) E and A (c) E and F (d) C and D

**385.** Consider a question and tqro statements : Question : Is always greater than Statement - I : x > y Statement - II : x 2 + > 2xy Which one Of the following is correct in respect Of the question and the State merits? [CDS (I) Paper 26 Q44]

(a) Statement - I alone is required to answer the question

(b) Statement - II alone is required to answer the question

(c) Both Statemen and Statement - II are required to answer the question

(d) Neither Statement - I nor State- ment - Il is required to answer the qu estion P.T.O.

**386.** If tane + sece 3, then What is the value of 3 tan8+ 9sece? (b) 17 [CDS (I) Paper 26 Q56]

(a) 120'

(b) 135'

(c) 150* 52. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of SS. What is

(d) following? aeroplane is Observed to be 56. If tane + sece 3, then What is the value of 3 tan8+ 9sece?

**387.** If O, where then what is 'the value Of 12cot2 B + 3cosece? [CDS (I) Paper 26 Q59]

(a) (0+112 (b) (0+2)2 (c) p" +1)2 (d) (30+1)2

**388.** How much less amount is allocated to A and C together as compared to D and E together? [CDS (I) Paper 26 Q95]

(a) 0004 lakhs (b) 0005 lakhs (c) 6008 lakhs (d) 0010 lakhs 45K

**389.** What is (1 + tana tanß)2 + (tana - tanß)2 equal to ? [CDS (I) Paper 31 Q12]

(a) tanza tan2ß (b) sec2a sec2ß (c) tan2a cot2ß (d) sec2a tan2ß

**390.** What is the value of sin2e cos29 (s«2e + cosæ2e) equal to ? sinle [CDS (I) Paper 31 Q15]

(a) 15. What is the value of 18. From a point X on a bridge across a sin29 cos2e

(b) 2h

(c) Htana*tanß

(d) h cota*cOtß

**391.** If 64 64 - 16 where O O - , 2 then what is the value of + coto ? [CDS (I) Paper 31 Q16]

(a) 15. What is the value of 18. From a point X on a bridge across a sin29 cos2e

(b) 2h

(c) Htana*tanß

(d) h cota*cOtß

**392.** If cosece - Cote - m and sece - tane n, then what is cosece + sece qual to ? (d) [CDS (I) Paper 31 Q17]

(a) 15. What is the value of 18. From a point X on a bridge across a sin29 cos2e

(b) 2h

(c) Htana*tanß

(d) h cota*cOtß

**393.** A person purchases one kg of tea powder from each of the four places A, B, C, D at the rate per 1 kg, 2 kg, 4 kg, 5 kg. If on an average he purchased x kg of tea powder per 1000, then what is the approximate value of x ? 9 9 [CDS (I) Paper 31 Q23]

(a) 195 (b) 2-00 (c) 205 (d) 2*10

**394.** If A is odd, then B is odd Select the answer using the code given below : [CDS (I) Paper 31 Q11]

(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 1 730 1 65

**395.** A 4-digit number N has exactly 15 distinct divisors. What is the total number of distinct divisors of ? [CDS (I) Paper 31 Q31]

(a) 16 (b) 30 (c) 45 (d) 225

*— SPACE FOR ROUGH WORK —*

**396.** If the LCMisx (a 2-digit number), then how many values of x are possible ? [CDS (I) Paper 31 Q10]
(a) Only one (b) Only two (c) Only three (d) More than three 13

**397.** What is equal to ? 2 cos30 - cose [CDS (I) Paper 31 Q40]
(a) sun29 (b) cos2e (c) cot9 (d) tano A -

**398.** A trolley with two wheels one metre apan is moved clockwise on the circular track around a ground with radius 50 m (described by right wheel). If the size of each wheel is of 1 foot radius and the right wheel turns 1000 times, how many times will the other wheel tum ? [CDS (I) Paper 31 Q44]
(a) 1010 (b) 1015 (c) 1020 (d) 1025

**399.** A vertical pole of length 80m is situated on the horizontal plane. The base of the pole is at P. There are two points A and B such that p, A, B are on the same straight line. Let the angles of elevation of top of the pole from A and B be a and (a>ß) respectively. If PA 64 m and AB 36 m, then what is (a +ß) equal to ? [CDS (I) Paper 31 Q46]
(a) 600 (b) 900 (c) 1200 (d) 1350 15 A -

**400.** If LBAC - then what is sine + cos9 equal to ? 71 65 73 65 79 (d) 65 [CDS (I) Paper 31 Q56]
(a) I and 11 only (b) 11 and Ill only
(c) I and Ill only (d) 1, 11 and 111 Consider the following for the next two

**401.** If BD is the perpendicular on the side AC, then what is the length of BD ? 1008 65 (b) cm 65 A - [CDS (I) Paper 31 Q57]
(a) I and 11 only (b) 11 and Ill only
(c) I and Ill only (d) 1, 11 and 111 Consider the following for the next two

**402.** A question is given followed by two statements I and [CDS (I) Paper 31 Q61]
(a) 5 cm (b) 6 cm 59. What is PM equal to ?
(c) IO cm (d) 12 cm

**403.** XIZ is a 3-digit number, where X, Y, Z are distinct non-zero digits. The dif- ference between the two 3-digit numbers XYZ and YXZ is [CDS (I) Paper 31 Q89]
(a) 121 (b) 120
(c) 119 (d) None of the above 91. What is the HCF 0f236-1 and 245-1 ?

**404.** How many possible values exist for the sum [CDS (I) Paper 31 Q90]
(a) 121 (b) 120
(c) 119 (d) None of the above 91. What is the HCF 0f236-1 and 245-1 ?

**405.** How many numbers of the form 2n - 1 and less than 2000 are prime ? 41 [CDS (I) Paper 31 Q97]
(a) 1100
(b) 1200
(c) 1300
(d) 1400 96. The number 199 can be written as m2 - n2, where m, n are natural numbers

**406.** 66. (Global Capability Centre) ? [CDS (I) Paper 32 Q64]
(a) (Honeywell)
(b) (Citibank)
(c) (Texas Instruments)
(d) (SAP Labs India) (GVC) (CAD) 1, Gvc (c) (Currency depreciation) (net exports) t? 1. (d) qi 1, 4t12 CEBN-s-GKL (20-

**407.** Moidam-The Mound-Burial system of the Ahom dynasty was inscribed in the UNESCO World Heritage lisu The Government Of India conferred the status of classical language to Pali, Assamese, Bengali and Maithili. Project PARI is an initiative Of Ministry of Education Select the answer using the code given below : [CDS (I) Paper 32 Q2]
(a) 2 only (b) I only (c) 2 and 3 only (d) 1,2and3

**408.** Which of the above are correct? [CDS (I) Paper 33 Q11]
(a) I and 11 only (b) 11 and Ill only (c) I and Ill only (d) 1, 11 and 111

**409.** What is c2 + bc - a2 - ab equal to? [CDS (I) Paper 33 Q21]
(a) 12 21. What is c2 + bc - a2 - ab equal to?
(b) -1/2 23. 24. 25. If x -2+21/2 +23/2 x2 - 4x - IO equal to? If equal to ?
(c) 2p
(d) then what is then what is x 22. If ab ba , then what is equal to?

**410.** If ab ba , then what is equal to? (d) ab 9 n. If x-2+21/2 +23/2 x2 - 4x - IO equal to? [CDS (I) Paper 33 Q22]
(a) 12 21. What is c2 + bc - a2 - ab equal to?
(b) -1/2 23. 24. 25. If x -2+21/2 +23/2 x2 - 4x - IO equal to? If equal to ?
(c) 2p
(d) then what is then what is x 22. If ab ba , then what is equal to?

*— SPACE FOR ROUGH WORK —*

**411.** If equal to ? 2p 2p then what is then what is x a - b 2 a+b x + b O, then what [CDS (I) Paper 33 Q24]
(a) 12 21. What is c2 + bc - a2 - ab equal to?
(b) -1/2 23. 24. 25. If x -2+21/2 +23/2 x2 - 4x - IO equal to? If equal to ?
(c) 2p
(d) then what is then what is x 22. If ab ba , then what is equal to?

**412.** If 2 tane=sec29 - 2, where R 12, then what is cote equal to? (c) v'ä-l (d) 0+2 [CDS (I) Paper 33 Q33]
(a) 3/2 5/3 33. If 2 tane=sec29 - 2, where R 12, then what is cote equal to?
(b) v'ä-l
(c) 0+2 I - sin 34. What is
(d) mane

**413.** If x4 + y4 14x2 , then consider the I. loglO(x 2 + ) loglo x *loglo y t 210g10 2 [CDS (I) Paper 33 Q38]
(a) 11 only (b) Both I and 11
(c) Neither I nor 11 39. Which Of the is/ are the Of (d) Both I and 11

**414.** loglO(x2 - ya ) - loglo x + loglo y Which of the above is/are correct? [CDS (I) Paper 33 Q11]
(a) I only (b) 11 only (c) Both 1 and 11 (d) Neither I nor 11

**415.** 13y) n. (x+19y) Select the correct answer using the code given below. [CDS (I) Paper 33 Q1]
(a) I only (b) 11 only (c) Both 1 and 11 (d) Neither I nor 11

**416.** What is +3 equal to? (b) 1/2 For the tvo items : Let psin2 +qcos2 a qsin2 pcos2 ß=n; p m, n and q m. n. [CDS (I) Paper 33 Q40]
(a) 11 only (b) Both I and 11
(c) Neither I nor 11 39. Which Of the is/ are the Of (d) Both I and 11

**417.** If a and are complementary angles, then which one Of the following is [CDS (I) Paper 33 Q42]
(a) mn - I O For the following two
(b) 2/3 For the following tvo
(c) 1 P.T.O. For the following two
(d) 2/3 sina 4 u sin # cosa cosß 9 45. What is tan 2 equal to? 46. What is tan 2 ß equal to? 1/3 For the following tvo

**418.** What is (p sine + qcos9)equal to? [CDS (I) Paper 33 Q43]
(a) mn - I O For the following two
(b) 2/3 For the following tvo
(c) 1 P.T.O. For the following two
(d) 2/3 sina 4 u sin # cosa cosß 9 45. What is tan 2 equal to? 46. What is tan 2 ß equal to? 1/3 For the following tvo

**419.** Ifp+q+r=0, then what is equal to? (c) 22 (d) 23 [CDS (I) Paper 34 Q1]
(a) -23 (b) -17 (c) 17 (d) 23 which - 5k + 4

**420.** What is the value of k for which (k2 - 5k + 4)x2 + (k2 - 3k - 4)x+ (k2 - 4k) 0 is an identity ? 152 [CDS (I) Paper 34 Q2]
(a) -23 (b) -17 (c) 17 (d) 23 which - 5k + 4

**421.** If then what is the value of a4 + b4 + ca equal to ? [CDS (I) Paper 34 Q3]
(a) a2b2 + + c2a2 (b) 2(a2b2 c2a2) (c) 3(a2b2+ Fc2 + c2a2) (d) 4(a2b2 b2c2 + c2a2) 3

**422.** If - -L - + , then what is pq x + p x equal to ? Fi¯i ¯ - q x - p x - q [CDS (I) Paper 34 Q4]
(a) -23 (b) -17 (c) 17 (d) 23 which - 5k + 4

**423.** If x varies directly as y, then which of the following is/ are correct ? I. (x2 + y2) varies directly as ya. varies directly as y. Select the answer using the code given below : [CDS (I) Paper 34 Q13]
(a) I only (b) 11 only (c) Both land 11 (d) Neither I nor 11

**424.** What is the sum of all the divisors of 256 ? [CDS (I) Paper 34 Q27]
(a) 511 (b) 510 (c) 256 (d) 255 A -

**425.** If m and n are relatively prime, then m x n is always even. Which of the statements given above is/are correct ? [CDS (I) Paper 34 Q11]
(a) I only (b) 11 only (c) Both land 11 (d) Neither I nor 11

**426.** If and then what is log10(384) equal to ? (b) p - q+r (d) p+q - r [CDS (I) Paper 34 Q29]
(a) I only (b) 11 only
(c) Both land 11 (d) Neither I nor 11 29. If and then what is log10

**427.** What is the digit in the unit place of the number 11 x 33 x 55 x 77 x 99 ? 13 [CDS (I) Paper 34 Q30]
(a) I only (b) 11 only
(c) Both land 11 (d) Neither I nor 11 29. If and then what is log10

*— SPACE FOR ROUGH WORK —*

**428.** The sum of the deviations of a set of n numbers Xl, x2, x3 , x meas- ured from 15 is - 90 and the sum of the deviations of the same numbers measured from - 3 is [CDS (I) Paper 34 Q39]

(a) 3-25 (b) 3-50

(c) 3-75 (d) 4-25 40. The numbers 59, 53, 51, 43, 36,

**429.** If 3sine + 4cos9 - 5, then what is a value of 4tan0 + 3cot0 ? [CDS (I) Paper 34 Q43]

(a) 3-25 (b) 3-50

(c) 3-75 (d) 4-25 40. The numbers 59, 53, 51, 43, 36,

**430.** Select the answer using the code given below : [CDS (I) Paper 34 Q11]

(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

**431.** ZPAD+ ZPCB Select the answer using the code given below : [CDS (I) Paper 34 Q11]

(a) I only (b) 11 only (c) Both landll (d) Neither I nor 11 A -

**432.** What is the value of q ? next two (02) items that For the follow : 16 + [CDS (I) Paper 34 Q82]

(a) 487 (b) 483 (c) 481 (d) 480 85. What

**433.** What is 65 8 61 8 59 For the follow : -257-0 equal to ? equal to ? next two (02) items that and y = Let x 39 [CDS (I) Paper 34 Q84]

(a) 487 (b) 483 (c) 481 (d) 480 85. What

**434.** What is a value of (cosecO + secO) ? 35 12 25 12 For the next two (02) items that follow : cosece - sino = p' and seco - cose = q3 [CDS (I) Paper 34 Q92]

(a) 1/2 (b) 1/√2 (c) √3/2 (d) 1

**435.** What is cot2ß equal to ? p For the next two (02) items that follow : p + q coto = 3cosec0 and q - p cote 2cosec9 [CDS (I) Paper 34 Q96]

(a) 96. What is cot2ß equal to ? p For the next two (b) 10

(c) 13 For the next two (d) 2pq - -

**436.** What is p2 + q2 equal to ? (d) 13 (c) 10 43 [CDS (I) Paper 34 Q97]

(a) 96. What is cot2ß equal to ? p For the next two (b) 10

(c) 13 For the next two (d) 2pq - -

**437.** Which one Of the following is correct ? [CDS (I) Paper 34 Q99]

(a) 96. What is cot2ß equal to ? p For the next two (b) 10

(c) 13 For the next two (d) 2pq - -

## SECTION : QUADRATIC EQUATIONS & PROGRESSIONS (15 QUESTIONS)

**438.** B starts with a wrong value Of q and gets the roots as 2 - 9. What the correct roots of the equation ? [CDS (I) Paper 7 Q6]

(a) 3 and - 4 (b) - 3 and 4 (c) - 3 and 4 (d) 3 and 4 64. of

**439.** What is the value Of 1001xlOOlxiOOl+0999x0999xo*999 1001x x0999+0999xo-999 If p(x) is a common multiple of degree 6 Of the (x) = x' + x2 - x - I and g(x) - - x2 + X - I then which Of the following is correct? [CDS (I) Paper 7 Q62]

(a) p (b) 3 and - 4 (c) - 3 and - 4 (d) - 3 and 4

**440.** The roots of the equation are always unequal and real, Which Of the above statements is/ are correct? [CDS (I) Paper 10 Q2]

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2 and ß are the roots or the

**441.** If the sum of the roots of ax + bx+c - O is equal to the sum of the squares of their reciprocals, then which one Of the following relations is correct? [CDS (I) Paper 15 Q16]

(a) bc 2 = 2a2c (b) ac2+ 2b 2 a (c) ab2+ bc 2' a 2c (d) a2+b2+c2=I

**442.** If p and q are the roots ofx2+px+q=0, then which one of the following is correct ? (b) p -loniy = - 2 or O = - 2 Ohly [CDS (I) Paper 16 Q20]

(a) p -loniy 20. 6 If p and q are the roots of x2+px+q = 0, then which one of the following is correct ? remaining work ?

(b) 6 hours

(c) 12 hours

(d) hours

**443.** If and are the roots of the equation 19, ax2 + bx + c = O, then the value of 1 au + b aß + b bc (b) 20 ab 1 (d) abc Consider the following statements in respect of three a-digit numbers xyz, yzx and ZXY : [CDS (I) Paper 19 Q18]

(a) ab 1

(b) abc equation 19. value of 20. zxr - 1, then aq ¯ ¯ P cP - q is equal to

(c) abc

(d) xyz

*— SPACE FOR ROUGH WORK —*

**444.** Which one of the following equations does not have real roots ? (b) 2r2+10x-1-o (d) *4x2 + A - [CDS (I) Paper 25 Q9]
(a) 29, 17, 37
(b) 31, 17, 47
(c) 19, 37, 13
(d) 23, 31, 37 2. Which one of the following is a factor of the

**445.** If p and q (p > q) are the roots of the equation xa - 60x + 899 = O, then which one of the following is correct ? -24+27=0 (c) 2p - q 30 0 (d) 3p - 2q 43 0 A - [CDS (I) Paper 25 Q18]
(a) only
(b) k>2v5 only
(c) k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is
(d) I - aßy

**446.** If the roots of the equation x2 - 4x - O are real, then what is the minimum value of N? [CDS (I) Paper 25 Q19]
(a) 0-1
(b) 0-01
(c) 0-001
(d) 0-0001

**447.** If one Of the roots of the equation is 4, then what is the sum of the squares of the roots ? 15 2 2 19 2 2 [CDS (I) Paper 25 Q31]
(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2 only when Ware correct ? 28. Let work done by

**448.** Let p, q be the roots of the equation x2 + mx - n 0 and m, n be the roots of the equation .r2 + px - q = O (m, n, p, g are non-zero numbers). Which of the following statements is/ are correct ? L m(m + n) = IL p+q-l (c) (d) 5 Select the answer using the code given below : [CDS (I) Paper 31 Q10]
(a) I only
(b) 11 only
(c) Both land 11
(d) Neither I nor 11

**449.** If p and q are the roots of the equation x2 - sin29x - cos2B 0, then what is the minimum value of p2 + q2 ? 3 2 [CDS (I) Paper 31 Q20]
(a) nM - k
(b) n
(c) 81
(d) 105

**450.** If are the roots Of this equation? 2' 2b [ P.T.O. [CDS (I) Paper 33 Q25]
(a) 12 21. What is c2 + bc - a2 - ab equal to?
(b) -1/2 23. 24. 25. If x -2+21/2 +23/2 x2 - 4x - IO equal to? If equal to ?
(c) 2p
(d) then what is then what is x 22. If ab ba , then what is equal to?

**451.** If a and are the roots Of the equation then what is equal to? [CDS (I) Paper 33 Q37]
(a) 16
(b) 25
(c) 36
(d) 49 1 P.T.O.

**452.** ABC is a triangle right-angled at C. A semicircle is drawn on AB as diameter. Let P be any point on AC produced such that AP = AB IO cm. Further, B and P are joined. If BC 8 cm, then what is BP equal to ? 15 [CDS (I) Paper 34 Q36]
(a) 443 cm
(b) 9 cm
(c) I Ocm
(d) cm

## SECTION : TRIGONOMETRY & HEIGHTS AND DISTANCES (51 QUESTIONS)

**453.** If sin x cos x = 1/2, then what is the value of sin x - cos x ? nt lat of Iftan2 y cosec2 x - 1 tana y, then which 81. one of the following is correct ? (b) x = 2y (c) y - 2x [CDS (I) Paper 1 Q80]
(a) The angle bisector of angle A
(b) The angle trisector Of angle A
(c) bisector of angle A
(d) None of the above What is log

**454.** 22 COS x cos x 2 , which one I + cosec x cosecx - of the following is one Of the values [CDS (I) Paper 1 Q83]
(a) The angle bisector of angle A
(b) The angle trisector Of angle A
(c) bisector of angle A
(d) None of the above What is log

**455.** sin x cos x 1/2 t, sin x - cos x 80. tan2 y coseca x - 1 = tan2 y t, 81. (d) x - y = 10 [CDS (I) Paper 1 Q84]
(a) x - y = 10 78. 79. 80. 81. 82. 83. 84. sin x cos x 1/2 t, sin x - cos x DE t
(b) *rut A log IO
(c) Tt/2
(d) Tt/3

**456.** 150m 300 I 450 [CDS (I) Paper 1 Q89]
(a) 59*4 m
(b) 61 *4m
(c) 62*4 m
(d) 63*4 m 90. w, (c) 2vif 24 91. sin x + sin y a cos x + cos y = b t, sinx siny+cosx (a) (b) (c) 25 a + b - ab

**457.** 150m gr 300 I 450 [CDS (I) Paper 1 Q91]
(a) 59*4 m
(b) 61 *4m
(c) 62*4 m
(d) 63*4 m (a) (b) (c) (d) 24 sin x + sin y - a cos x + Cosy = b t, sinx siny+cosx (a) (b) (c) (d) a + b - ab 2

*— SPACE FOR ROUGH WORK —*

**458.** Which one of the following is correct ? [CDS (I) Paper 3 Q11]
(a) There is only one g with 00 < < 900 such that sin = a, where a is a real number
(b) There is more than one O with 00 < e 900 such that sin = a, where a is a real number - A 4
(c) There is no 9 with 00 < 900 such that sin a. where a is a real
(d) There are exactly two Os with < 900 such that sin = a, where a is a real number

**459.** If sinO+cosO=i, then what is the value of sine dose? 2 (d) 1/2 I + sin a [CDS (I) Paper 4 Q46]
(a) 1/2 47. (b) 60 (c) 45 (d) 30

**460.** What is equal to? I - sin0 (b) secG+ tane (c) cosece + cot é (d) cosec - c Ot [CDS (I) Paper 4 Q47]
(a) 1/2 47. (b) 60 (c) 45 (d) 30

**461.** The shadow Of a tower is 15 m when the Sun's altituqe is [CDS (I) Paper 4 Q48]
(a) 1/2 47. (b) 60 (c) 45 (d) 30

**462.** sm t? I - sine [CDS (I) Paper 4 Q49]
(a) 9 - tan 9
(b) sece+ tan 8
(c) cosece + cot 9
(d) cosecO - cot 9 30' t, 15 m 60* t, (d) 100 (b) (c) 8905 83 192 33 88-55 8755 89-55 26A IS (c)

**463.** 5 mm [CDS (I) Paper 5 Q94]
(a) 22 m
(b) 20 m
(c) 18 m 95. qt«rq (cm2 t ? 16Jä -8T 16Jä - (a) (b) (c)
(d) 16 NG 27 - ' g 96. (a) 3-5 km (b) km (c) 5*5 km (d) 6-5 km cot 19 cot 200 cot 700 cot 75' '

**464.** sin cos (9-20) 6 0 - cos e - cos I - sin2 B cos2 e 2-5 kmph 3aqå 3-25 kmph t I 4 36 [CDS (I) Paper 5 Q97]
(a) 16 cm (b) 20 em
(c) 25 cm (d) 30 cm ABC !3T3-å AB AC qi D 3k E BC 3 ADE 3 cm2 I EC ?

**465.** 0 - cos e - cos [CDS (I) Paper 5 Q93]
(a) 16 cm (b) 20 em
(c) 25 cm (d) 30 cm ABC !3T3-å AB AC qi D 3k E BC 3 ADE 3 cm2 I EC ?

**466.** What is the length of the perpendicular drawn from the centre Of circle of radius r On the chord of length r ? (c) r/2 If a and ß are complementary angles, then sin a Cosa - i what is coseca * cosec ß sin cosß equal to ? (d) None of the abo ve Consider the following statements : [CDS (I) Paper 6 Q87]
(a) 280
(b) 52 What is the length of the perpendicular drawn from the centre Of circle of radius r On the chord of length r ?
(c) r/2 If a and ß are complementary angles, then sin a Cosa - i what is coseca * cosec ß sin cosß equal to ?
(d) None of the abo ve Consider the following statements : 88. 89. 90. 91. Which one of the following is a prime number ?

**467.** 87. (b) 280 (d) 52 What is the length of the perpendicular drawn from the centre Of circle of radius r On the chord of length r ? (c) r/2 If a and ß are complementary angles, then sin a Cosa - i what is coseca * cosec ß sin cosß equal to ? (d) None of the abo ve Consider the following statements : [CDS (I) Paper 6 Q85]
(a) 280
(b) 52 What is the length of the perpendicular drawn from the centre Of circle of radius r On the chord of length r ?
(c) r/2 If a and ß are complementary angles, then sin a Cosa - i what is coseca * cosec ß sin cosß equal to ?
(d) None of the abo ve Consider the following statements : 88. 89. 90. 91. Which one of the following is a prime number ?

**468.** There is only one Value Of x in the first quadrant that satisfies sin x - cos x = [CDS (I) Paper 7 Q2]
(a) 64 minutes
(b) 80 minutes
(c) 96 minutes
(d) 120 minutes The pair of numbers which are relatively prime to each other is :

**469.** Consider the following : cot 300 + 1 = 2 (cos 300+1) cot 300-1 [CDS (I) Paper 7 Q98]
(a) I only (b) 2 only
(c) Both land2 (d) Neither nor 2 If 3 sin x + 5 cos x = 5, then.what is the value of

**470.** If tan O 3/4 and O is acute, then what is sin 0 equal to ? [CDS (I) Paper 8 Q47]
(a) ac I If f (b) -20 (c) 20 (d) 12

*— SPACE FOR ROUGH WORK —*

**471.** m, then the height Of the tower is close to [CDS (I) Paper 8 Q62]
(a) 86.6m
(b) 84.6 m
(c) 82.6 m
(d) 80.6 m 16 In a triangle ABC, LBCA = 600 and AB2 = BC2 + CA2 + X. What is X equal (a) (b) (c) (d) Zero

**472.** A - sin A What is gin3 A €9) - cos A see A sec A cos A None Of the ain A and is an angle. A equal if Sih following correct ? cos B COS * cos COS 2 sin O cos O 2 sin O' sec B tan b one Of A - sin A 42 [CDS (I) Paper 9 Q44]
(a) 1 (b) √3 (c) 1/√3 (d) 2

**473.** What is equal to? cotx tan x [CDS (I) Paper 12 Q56]
(a) sin x (b) cos x (c) tan x (d) cot x 18

**474.** What is (sin x - cos equal to? sm x cos [CDS (I) Paper 12 Q59]
(a) 900
(b) 13 sin x - cos x + I What is equal to? sin X + cos x - I sin x - I COS X sin x + I
(c) sin x - cos x + I sin X +1
(d) COS x + 1 59. 60. What is

**475.** m 300m 300 Jfä 500 600 4Tä 900 (d) sin x - cos x + I sin X 4-cosx - I sin X - I [CDS (I) Paper 12 Q57]
(a) sin x - cos x + I sin X 4-cosx - I cos x - cos y sin X - I (b) COS X sin x + I cos x y COS X sin x - I sin x - cos y
(c) COS X + I sin x + I (d) sin 2 x - sin y cosx+ i IA 59. 60. 21

**476.** COS < COS Which Of the above Stater-nents is / are correct? [CDS (I) Paper 12 Q2]
(a) I Only (b) 2 only (c) Both 1 and 2 (d) Neither I nor 2

**477.** DCM + ABM ABCD e I 61. (l t? (e) sin x (d) cos x (cosecx - sin x) (sec x - cos x) (tan x + cot* ) [CDS (I) Paper 12 Q2]
(a) cos x (b) sin X + cos x
(c) sin x - cos x 63. * l. sin r > sin 1 2. cos l' cos I (d) 4q 66. 67.

**478.** If x=acosO and y=bcot9, then + by-I) is equal to (c) tan29 (d) sin2e COS 9 38. is equal to I - sine tang - I tang +1 1 + sine (b) coso tano + I tang - I (d) sin 6 [CDS (I) Paper 14 Q37]
(a) tan29 (b) sin2e COS 9 38. is equal to I - sine tang - I tang +1 1 + sine
(c) coso tano + I tang - I (d) sin 6 39. If tan

**479.** If where B - 2 tan(x + + 20) - x) Otan(SO - = I, then the value of x is equal to [CDS (I) Paper 14 Q39]
(a) tan29 (b) sin2e COS 9 38. is equal to I - sine tang - I tang +1 1 + sine
(c) coso tano + I tang - I (d) sin 6 39. If tan

**480.** Two observers are stationed due north Of a tower (of height x metre) at a distance y metre from each other. The angles Of elevation of the tower observed by them are 30' and 45' respectively. Then x y is equal to [CDS (I) Paper 15 Q65]
(a) 2 -Be , thenC+2B
(b) 0-1 0+1 cos26 - 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e
(c) only
(d) 2 only

**481.** The value of -loglo g10 2 32 5 is If tane + cote , where O < < then 2 sin + cos 9 is equal to 0-1 (c) 2 / 54A [CDS (I) Paper 15 Q96]
(a) AM < Median (b) GM Median
(c) AM = Me dian Suppose for Which one of the corre ct? (d) GM>1/6

**482.** If sin x + sin2x=l, then what is the value of cossx + 2cos6x + cos x ? [CDS (I) Paper 16 Q72]
(a) 4275
(b) 4350
(c) 4375
(d) 4450 68. What is the area of a triangle with sides Of length 12 cm, 13 cm and 5 cm ?

**483.** If tan2x+ = 2 and 00 < x < 900, tan x then what is the value of x ? [CDS (I) Paper 16 Q81]
(a) 150 (b) 300 (c) 450 (d) 600

**484.** Ifthe length of the shadow of a tower is equal to its height, then what is the Sun's altitude at that time ? [CDS (I) Paper 16 Q82]
(a) 150 (b) 300 (c) 450 (d) 600

**485.** Consider the following : COS 750 sin 150 cos 350 2. sin 550 sin 800 3. cos sin 120 cos 180 cos 780 sin 720 sin 1 10 cos 790 cos 280 cosec 620 = 1 - sin 590 sec 310=0 Which Of the above are correct ? [CDS (I) Paper 16 Q83]
(a) 1 and 2 only (b) 2 and 3 only (c) I and 3 only (d) 1, 2 and 3 - A 24

*— SPACE FOR ROUGH WORK —*

**486.** What isthevalueof2-!-E cos730 eos71P Sin 170 C - [CDS (I) Paper 21 Q35]
(a) Its lia ixuide O
(b) Its lies on the triülgle
(c) It 30. Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ?
(d) AB<RX

**487.** What is the least value of (25 cosec2x + sec2x) ? [CDS (I) Paper 22 Q52]
(a) 40 (b) 36
(c) 26 (d) 24 and If a cot n9, then which one of following is correct ?

**488.** cosa + cosß 2 3m sina sinß- O t, Osa s 9C, O S ß S 909 cosm -cos2ß t? 56. sece+cosB- - , os0S900 t, sin 2 e t? 2 3 57. (1+cot8 t sece) 4 3 2 [CDS (I) Paper 24 Q54]
(a) 11 (b) 22
(c) 77 (d) 99 k sece 59. 60' 60. 60* 45'

**489.** Consider the following : 2. sin4Ø+cos49=l+2sin20cos20 15 15 Which Of the above are identities ? [CDS (I) Paper 25 Q53]
(a) I and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3 M. What is the value of sin 240 sin 660 - cos240 cos 66 + tan 240 tan 660 - cot 240 cot 660 ?

**490.** What is cos9 - 2cos3e' equal to ? [CDS (I) Paper 25 Q58]
(a) sin 9 (b) cose (c) tan 0 (d) cot O

**491.** What is the value of the following? 2 sin 680 2cot 150 cos 220 5 tan 750 3 tan20' tan4cr tan 450 tan 5 The perpendicular dropped from a vertex 12. of a right-angled triangle the hypotenuse divides it into two segments Of lengths 9 units and 16 units respectively. What is the length Of the perpe n dicular? [CDS (I) Paper 26 Q11]
(a) 6 units (b) 8 units (c) IO units (d) 12 units

**492.** The angle of elevation Of the top Of a tower of height x metre from a point on the ground is found to be 60*. By going y metre away from that point, it becomes [CDS (I) Paper 26 Q54]
(a) 120'
(b) 135'
(c) 150* 52. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of SS. What is
(d) following? aeroplane is Observed to be 56. If tane + sece 3, then What is the value of 3 tan8+ 9sece?

**493.** 53. tan (3A) (A - 220), SA W [CDS Paper 30 Q51]
(a) 250 (b) 270
(c) 280 (d) 300 sin e - Cos + I z p sec 9 + q tan O, sin cos 54. :

**494.** If o 2' + P) -.13 and tan(a - ± , then what is tana*cot2ß equal to ? [CDS (I) Paper 31 Q14]
(a) 15. What is the value of 18. From a point X on a bridge across a sin29 cos2e
(b) 2h
(c) Htana*tanß
(d) h cota*cOtß

**495.** Two towers A and B of height 23 m and 11 m respectively, stand 9 m apart. A straight rod is joined to the two tops of the towers. A monkey sitting on the top of A, climbs the rod to reach the top of B. If the monkey takes 5 minutes to reach the other end, what is the average speed of the monkey ? [CDS (I) Paper 31 Q42]
(a) IO mimin (b) 5 mimin (c) 10 cm/sec (d) 5 cm/sec

**496.** What is (sece-tan9) to? (b) mane (d) sine + cos B I - sin I + sin 9 6, then What is as. If cote coec2e - sec2 e coxc29+sec e equal to? [CDS (I) Paper 33 Q34]
(a) 1/2 (b) 1/3
(c) 2/3 (d) 3/4 difference between the two acute

**497.** What is p2q2(p2 + q2 + 3) equal to? For the following two (02) items : (d) 2/3 sina 4 u sin # cosa cosß 9 [CDS (I) Paper 33 Q44]
(a) mn - I O For the following two
(b) 2/3 For the following tvo
(c) 1 P.T.O. For the following two
(d) 2/3 sina 4 u sin # cosa cosß 9 45. What is tan 2 equal to? 46. What is tan 2 ß equal to? 1/3 For the following tvo

*— SPACE FOR ROUGH WORK —*

**498.** What is tan 2 equal to? [CDS (I) Paper 33 Q45]
(a) mn - I O For the following two
(b) 2/3 For the following tvo
(c) 1 P.T.O. For the following two
(d) 2/3 sina 4 u sin # cosa cosß 9 45. What is tan 2 equal to? 46. What is tan 2 ß equal to? 1/3 For the following tvo

**499.** What is tan 2 ß equal to? 1/3 For the following tvo (02) items : I + sin e COS 9 [CDS (I) Paper 33 Q46]
(a) mn - I O For the following two
(b) 2/3 For the following tvo
(c) 1 P.T.O. For the following two
(d) 2/3 sina 4 u sin # cosa cosß 9 45. What is tan 2 equal to? 46. What is tan 2 ß equal to? 1/3 For the following tvo

**500.** If cos I - sina cos O = 4, then which I + Sin one of the following is a value of (tan29 + cot29) ? 10 17 (b) (d) 17 [CDS (I) Paper 34 Q41]
(a) 3-25 (b) 3-50
(c) 3-75 (d) 4-25 40. The numbers 59, 53, 51, 43, 36,

**501.** For O < - , consider the following : I. (tan40 + + c0160) = sec29 cosec29 tan 0 + sin 0 = cor29(sec9 + 11. tan - sin O Which of the above is/are identities ? [CDS (I) Paper 34 Q42]
(a) I only (b) 11 only (c) Both land 11 (d) Neither I nor 11

**502.** At a point on level ground, the tangent of the angle of elevation of the top of a tower is found to be S. On walking 70 m towards the tower, the tangent of the angle of elevation of the top of the tower is found to be Y. What is the height of the tower ? [CDS (I) Paper 34 Q44]
(a) 3-25 (b) 3-50
(c) 3-75 (d) 4-25 40. The numbers 59, 53, 51, 43, 36,

**503.** What is tane equal to ? (29+3p) [CDS (I) Paper 34 Q98]
(a) 96. What is cot2ß equal to ? p For the next two (b) 10
(c) 13 For the next two (d) 2pq - -

## SECTION : GEOMETRY: LINES, ANGLES, TRIANGLES & POLYGONS (69 QUESTIONS)

**504.** Two sides Of a parallelogram are IO cm and IS cm. If the altitude corresponding to the side Of length 15 cm is 5 crn, then what is the altitude to the side Of length [CDS (I) Paper 1 Q48]
(a) 1/64 What is the greatest number which divides 392, 486 and 627 so as to leave the same remainder in each case ?
(b) 47
(c) 43
(d) 34 A man walking at the rate 3 km.,'hr crosses a field diagorully in I minute. What is the area of the field ?

**505.** In the figure given above, O is the centre Of a circle circumscribing a quadrilateral ABCD. If AB BC and ZBAC 400, then what is Z A DC equal to ? [CDS (I) Paper 1 Q75]
(a) 500 (b) 600 (c) 700 (d) 800

**506.** Assertion [CDS (I) Paper 1 Q98]
(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true 98. Assertion

**507.** ABC is a triangle. Let D, E denote the mid points Of BC, CA respectively. Let AD and BE intersect at G Let O be a point on AD such that AO : OD 2 : [CDS (I) Paper 1 Q99]
(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true 98. Assertion

**508.** Staternent - I : I.Rt LMNbea triangle. Let P, Q be the mid- points of the sides LM, LN respectively. If PQ2&MP2+NQ2, then LMN is a fight angled triangle right angled at L. Statement - II : If in a triangle ABC, .4B2 > BC + CA2, then LACB is obtuse. Which of the following is correct in the light of the above statements ? [CDS (I) Paper 3 Q6]
(a) Both statement - I and statement 11 are correct and statement - 11 is the reason for statement - I
(b) Both statement - I and statement - 11 are corect and statemeht - 11 is not the reason for statement - I
(c) Statement - I is true, but state: ment - 11 is false
(d) Statement - I is false, but state - ment - 11 is true 2

*— SPACE FOR ROUGH WORK —*

**509.** What is tlE median of the ages Of minor children ? (b) 5 years (c) 7 years (d) Cannot determined due to in- sufficient data [CDS (I) Paper 3 Q8]

(a) 3 years (b) 4 years

(c) 5 years (d) 6 years 8. What is the median of the ages of minor children ?

**510.** The median BD of ghe triangle ABC meets at D. If then which one Of the following is correct ? [CDS (I) Paper 3 Q20]

(a) ZACB = I right angle (b) LBAC = I right angle (c) LABC I right angle (d) None of ttw abové - A 6

**511.** ABC is a right triangle with right angle at A. If the value of tan B = any.real k the length of the hypotenuse is of the form (b) 2k (d) 9k (c) 5k 10 [CDS (I) Paper 3 Q39]

(a) 4 days (b) 5 days (c) 6 days (d) days 33. If of

**512.** The perimeter of a triangle is greater than the sum of the lengths of its three medians. Which of the statemepts given above is/are correct ? [CDS (I) Paper 3 Q2]

(a) only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2 - A 16

**513.** Consider the follOW'ing statements * If two mangles are equi- angular, then they are similar. [CDS (I) Paper 4 Q88]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87* If

**514.** ABCD is a quadrilateral, the sides of Which touch a circle. Which one Of the following is correct? [CDS (I) Paper 4 Q90]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87* If

**515.** What is the product of the two numbers ? [CDS (I) Paper 7 Q4]

(a) 360

(b) 480

(c) 486

(d) 512 Consider the following statements in respect Of an equilateral triangle ABC.

**516.** Let ABC be a triangle with AB = 3 Cm, AC = 5 cm. If AD is a median drawn from the vertex A to the side BC, then which one of the following is correct ? [CDS (I) Paper 8 Q71]

(a) AD is always greater than 4 cm but less than 5 cm (b) AD is always greater than 5 cm

(c) AD is always less than 4 cm (d) None of the above Consider the following statements

**517.** In the figure given above YZ is parallel to MN, xy is parallel to LM and XZ is parallel to LN. Then MY is * [CDS (I) Paper 8 Q72]

(a) the median of triangle LMN (b) the angular bisector of ZLMN

(c) perpendicular to LN (d) perpendicular bisector Of LN

**518.** 71. (b) 350 (c) 400 (d) 45 Let ABC be a triangle with AB = 3 Cm, AC = 5 cm. If AD is a median drawn from the vertex A to the side BC, then which one of the following is correct ? [CDS (I) Paper 8 Q69]

(a) AD is always greater than 4 cm but less than 5 cm (b) AD is always greater than 5 cm

(c) AD is always less than 4 cm (d) None of the above Consider the following statements

**519.** E is the midpoint of the median AD of a triangle ABC If BE produced meets the side AC at F, then CF is equal to [CDS (I) Paper 10 Q76]

(a) AC/3 (b) 2AC/3 (c) AC/2 (d) None of the above

**520.** If every side Of an equilateral triangle is doubled, then the area Of new triangle becomes k times the area Of the Old One. What is k equal to? 17 (d) . The 7 4 3 sides Of a triangle are in the ratio - : If its perimeter is 52 cm, then What is the length Of the smallest side? [CDS (I) Paper 12 Q20]

(a) 9 cm (b) I O Crn (c) 11 cm (d) 12 cm

**521.** ABCD is a parallelogram. p and R are the midpoints of DC and BC respectively. The line PR intersects the diagonal AC at Q The distance CQ Will be equal to [CDS (I) Paper 12 Q70]

(a) AC/4

(b) BD/3

(c) BD/4

(d) AC/3 Consider the following Statements in respect Of an equilateral triangle * The altitudes are congruent.

**522.** A circle of radius r is inscribed in a regular polygon with n sides (the circle touches all sides of the polygon). If the perimeter of the polygon is p, then the area of the polygon is (b) 128 (b) (d) 2 None of the above [CDS (I) Paper 14 Q85]

(a) 20 square t

(b) 30 square unit

(c) 40 square unit

(d) None of the above 84. Chord CD intersects the diameter AB of a circle at riW1t angle at a m

**523.** How many right angled triangles can formed by joining the vertices of a cuboid ? [CDS (I) Paper 14 Q95]

(a) 24 (b) 28 (c) 32 (d) None of the above

*— SPACE FOR ROUGH WORK —*

**524.** If a variable takes discrete values a + 4, a-3*5, a-2*5, a-3, a-2, a+O*5, a +5 and a - 08 where a > O, then the median of the data set is [CDS (I) Paper 14 Q98]

(a) 24 (b) a - 25 (c) 28 (d) a - 1 25

**525.** In a triangle ABC if A is equal to (c) 2 -B- , thenC+2B [CDS (I) Paper 15 Q67]

(a) 2 -Be , thenC+2B

(b) 0-1 0+1 cos26 - 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e

(c) only

(d) 2 only

**526.** ABC is an equilateral triangle and X, Y and Z are the points on BC, CA and AB respectively such that BX CY = AZ. Which Of the following is/ are correct? I. XYZ is an equilateral triangle. [CDS (I) Paper 15 Q88]

(a) 24 16 = then the In the figure given above, AD CD What is the value of Z CDB?

(b) 320

(c) 640 BC ABC is a triangle right angled at C as shown in the figure above. Which one of the following is correct?

(d) AQ2+AB2 2 pQ2 BP2

**527.** Triangle XYZ is similar to triangle Select the correct answer using thc code given below. only [CDS (I) Paper 15 Q2]

(a) 24 16 = then the In the figure given above, AD CD What is the value of Z CDB?

(b) 320

(c) 640 BC ABC is a triangle right angled at C as shown in the figure above. Which one of the following is correct?

(d) AQ2+AB2 2 pQ2 BP2

**528.** If a quadrilateral has an Inscribed then the sum of a pair Of opposite sides equals [CDS (I) Paper 16 Q53]

(a) Half the sum of the diagonals (b) Sum of the other pair of opposite sides

(c) Sum Of two adjacent sides (d) None of the above

**529.** In an equilateral triangle another equilateral triangle is drawn inside joining the mid-points Of the sides of given equilateral triangle and the process is continued up to 7 times. What is the ratio of area of fourth triangle to that Of seventh triangle ? [CDS (I) Paper 16 Q56]

(a) 256 : I (b) 128 : 1 (c) 64: 1 (d) 16:1

**530.** The sides of a triangle are given by c + a2 and (b + c) where a +122, a, b, c are positive. What is the area of the triangle equal to ? 2 a2b2 +b2c2 +c2a2 (b) 2 2 3(a2b2 +b2c2 +c2a2) (d) 2 [CDS (I) Paper 16 Q61]

(a) 367t cm3 (b) 187t cm3

(c) cm3 61. The sides of a triangle are given by c + a2 and (d) 2 2 3

**531.** What is the area of a triangle with sides Of length 12 cm, 13 cm and 5 cm ? [CDS (I) Paper 16 Q68]

(a) 30 crn2 (b) 35 cm2 (c) 40 cm2 (d) 42

**532.** A pole stands vertically inside a triangular park ABC. If the angle of elevation Of the top Of the pole from each corner of the park is same, then in the triangle ABC, the foot of the pole is at the [CDS (I) Paper 16 Q84]

(a) Centroid (b) Circumference (c) Incentre (d) Orthocentre

**533.** The difference between their interior angles is [CDS (I) Paper 19 Q4]

(a) ab 1

(b) abc equation 19. value of 20. zxr - 1, then aq ¯ ¯ P cP - q is equal to

(c) abc

(d) xyz

**534.** ABC is a tingle riÖt at B. If ABS 5 cm BC = What is of the p*iølar drawn from B the hypotenuse ? 243 (e) [CDS (I) Paper 21 Q48]

(a) 32%

(b) 30% Sl. If the Of h+ue of a trian* is, IO is tie of awh a riørt mgld ?

(c) cm2

(d) 50

**535.** If the angles of a triangle are 300 and 450 and the included side is + 1) cm, then what is the area of the triangle ? [CDS (I) Paper 22 Q67]

(a) +1) cm2 (b) + 3) cm2 (c) l) cm2 (d) 266+1) cm2

**536.** Then what is the ratio of the area of the triangle formed by the parts of the diagonals with a long side to the area of the triangle formed by the parts of diagonals with a short side ? (b) 4.3 (c) 16:9 [CDS (I) Paper 22 Q3]

(a) - *E (b) litre (c) litre , (d) - litre 10

**537.** ABCD is a parallelogram where AC and BD are the diagonals. If ZBAD 600, ZADB 900, then what is BD2 equal to ? [CDS (I) Paper 22 Q78]

(a) - AB (b) a AB2 (c) 1, 2,3 82. 83. 2 (d) 4

**538.** A line through the vertex A of a parallelogram ABCD meets DC in p and BC produced in Q. If P is the mid-point of DC, then which of the following Ware correct ? [CDS (I) Paper 22 Q79]

(a) - AB (b) a AB2 (c) 1, 2,3 82. 83. 2 (d) 4

*— SPACE FOR ROUGH WORK —*

**539.** If the diagonals Of a rhombus are X and y, then what is its area ? 2 (b) 4 (c) xy [CDS (I) Paper 22 Q81]
(a) - AB (b) a AB2 (c) 1, 2,3 82. 83. 2 (d) 4

**540.** The lengths of sides of a triangle are where If one of the angles is 900, then what are the minimum integral values of x, y, z respectively ? [CDS (I) Paper 22 Q82]
(a) - AB (b) a AB2 (c) 1, 2,3 82. 83. 2 (d) 4

**541.** What the maximum number of circum-circles that a triangle can have ? (d) Infinite [CDS (I) Paper 22 Q83]
(a) - AB (b) a AB2 (c) 1, 2,3 82. 83. 2 (d) 4

**542.** ABCD is a quadrilateral such that AD = DC CA = 20 units, BC 12 units and ZABC= [CDS (I) Paper 22 Q89]
(a) 7-5 m 86. What is the area of the largest square plate cut from a circular disk Of radius one unit ?
(b) 4 square units
(c) 245 square units
(d) r square units

**543.** What is the median Of the following data ? [CDS (I) Paper 24 Q93]
(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015?
(b) and E
(c) A, C and D only
(d) C D and E

**544.** If A, B and C are interior angles of a triangle ABC, then what is equal to ? 2 A+B+C (c) sin A+B+C (d) tan 4 In a triangle ABC, right-angled at B, AB + BC + cm and length of the hypotenuse is 20 cm. What is the value of tanA + tan C ? 4 2 (d) 20 17 the 17 [CDS (I) Paper 25 Q59]
(a) sin 9
(b) cose
(c) tan 0
(d) cot O 59. If A, B and C are interior angles of a triangle ABC, then what is equal to ? 2 A+B+C

**545.** The radius of circum-circle of a right- angled triangle is 10cm and the altitude drawn to the hypotenuse is 8 cm. What is the area of the triangle ? [CDS (I) Paper 25 Q61]
(a) cm2 (b) 80 cm2 (c) 100 cm2 (d) 120 cm2

**546.** ABC is a triangle right-angled at B with AB 8 cm and BC = 6 cm. It is made to revolve about its side BC. What is the approximate total surmce area of the cone so formed ? (take [CDS (I) Paper 25 Q74]
(a) 452 cm2 (b) 440 cm2 (c) 432 cm2 (d) 420 cm2

**547.** What is the length of the altitude of the triangle drawn from vertex A on BC ? [CDS (I) Paper 25 Q82]
(a) 10*05 cm2 (b) 10*15cm2
(c) 11-05 cm2 (d) 11*15cm2 Consider the following for the next three

**548.** If p, Q and R are the areas of sectors at A, B and C within the triangle respec- tively, then which of the following is/are correct ? l. P = Æcm2 [CDS (I) Paper 25 Q83]
(a) 10*05 cm2 (b) 10*15cm2
(c) 11-05 cm2 (d) 11*15cm2 Consider the following for the next three

**549.** If AD-5cm, BD-4cm and 6 cm, then what is AB equal to ? [CDS (I) Paper 25 Q85]
(a) 10*05 cm2 (b) 10*15cm2
(c) 11-05 cm2 (d) 11*15cm2 Consider the following for the next three

**550.** If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of the (b) [CDS (I) Paper 26 Q52]
(a) 120'
(b) 135'
(c) 150* 52. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of SS. What is
(d) following? aeroplane is Observed to be 56. If tane + sece 3, then What is the value of 3 tan8+ 9sece?

**551.** In , a right triangle ABC, BD is perpendicular on hypotenuse AC If AC 9 cm and AD 4 cm, then what is AB+ BC approximately equal to? [CDS (I) Paper 26 Q90]
(a) 12 cm (b) 12*2 cm (c) 12*4 cm (d) 126 cm

**552.** In a triangle ABG AD is the bisector of ZBAC If 12 em, IO cmand DC - 5 cm. then what is the perimeter of the triangle? [CDS (I) Paper 26 Q87]
(a) 30 cm (b) 31 cm (c) 33 cm (d) 35 cm

**553.** If p, q and r are the lengths (in cm) of the sides of a right-angled triangle, then (p - q - - r - - p - q) (d) 16 is always [CDS (I) Paper 31 Q32]
(a) Positive only (b) Negative only (c) Non-positive only (d) Non-negative only

*— SPACE FOR ROUGH WORK —*

**554.** The base of a right-angled triangle is times the height of triangle. If the area of the triangle is 54 square cm, then what is the perimeter of the triangle ? [CDS (I) Paper 31 Q49]

(a) 30 cm (b) 32 cm (c) 36 cm (d) 40 cm

**555.** What is the median of the distribu- tion ? [CDS (I) Paper 31 Q54]

(a) 44-9 (b) 45-5 (c) 45-9 (d) 46-3 19

**556.** What is the mode of the distribution ? [CDS (I) Paper 31 Q55]

(a) I and 11 only (b) 11 and Ill only

(c) I and Ill only (d) 1, 11 and 111 Consider the following for the next two

**557.** The ratio of sum of interior angles to sum of exterior angles of a regular polygon of n sides is [CDS (I) Paper 31 Q95]

(a) 1100

(b) 1200

(c) 1300

(d) 1400 96. The number 199 can be written as m2 - n2, where m, n are natural numbers

**558.** What is the measure of an interior angle of polygon ? [CDS (I) Paper 31 Q4]

(a) 1100 (b) 1200 (c) 1300 (d) 1400

**559.** In a right triangle ABC, 900 and AD is perpendicular to BC. If ECAD 600 and BC = 6 cm, then what is AB equal to ? [CDS (I) Paper 31 Q100]

(a) 1100

(b) 1200

(c) 1300

(d) 1400 96. The number 199 can be written as m2 - n2, where m, n are natural numbers

**560.** The angles in a right-angled triangle is - radian. One of the acute angles Of the triangle is [CDS (I) Paper 33 Q36]

(a) 600 (b) 57-50 (c) 52-50 (d) 47-50

**561.** Let AD be the altitude of a triangle ABC If and (BD - CD) r, then what is BC equal to? (b) (d) qr / p pr / q pq/r [CDS (I) Paper 33 Q61]

(a) 20%

(b) 40%

(c) None of the above 61. Let AD be the altitude of a triangle ABC If and

(d) I only

**562.** ABC is a triangle right angled at B. P is the midpoint Of AB and Q is the mid- point of BC. Consider the following : I. AQ units 11. units Which of the atxwe is/ are required to determine the area Of the triangle? [CDS (I) Paper 33 Q63]

(a) I only (b) 11 only (c) Both I and 11 (d) More information is needed

**563.** The quadrilateral AAY is a The area Of the quadrilateral AZXY is half Of the area Of the triangle Which Of the statements given above is/ are correct? [CDS (I) Paper 33 Q11]

(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

**564.** In a triangle ABG ZA 300, AB 7 cm and AC* 12 Ctn. What is the area of the triangle ABC ? [CDS (I) Paper 33 Q71]

(a) 21 cm (b) 210 cm2 (c) 42 cm (d) 420 cm2

**565.** ABC is a triangle right angled at B. D is a point on AC such that BD is perpendicular to AC If AB p and BC v'äp, then what is BD equal to? (b) p 12 (d) Op/4 [CDS (I) Paper 33 Q72]

(a) 15 200 73.

(b) 10

(c) 11

(d) 12 71. In a triangle ABG ZA 300, AB 7 cm and AC* 12 Ctn. What is the area of the triangle ABC ?

**566.** The difference between an interior angle and an exterior angle of a regular polygon is 1200. What is the number of sides of the polygon? 9 (b) 10 (c) 11 (d) 12 ,'45A [CDS (I) Paper 33 Q73]

(a) 15 200 73.

(b) 10

(c) 11

(d) 12 71. In a triangle ABG ZA 300, AB 7 cm and AC* 12 Ctn. What is the area of the triangle ABC ?

**567.** The numbers 59, 53, 51, 43, 36, (8x+ l), (x2+ l), 13, 12, 9, 7, 3 are written in descending order and their median is [CDS (I) Paper 34 Q40]

(a) 3-25 (b) 3-50

(c) 3-75 (d) 4-25 40. The numbers 59, 53, 51, 43, 36,

**568.** ABC is a triangle in which AB = AC and D is any point on BC. Which of the following statements is/ are correct ? L AB2-AD2=CDXBD IL BC2+ Select the answer using the code given [CDS (I) Paper 34 Q48]

(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

*— SPACE FOR ROUGH WORK —*

**569.** ABC is a triangle right-angled at C. Let p be the midpoint of BC. If AP cm and AB 20 cm, then what is the perimeter of the triangle ABC? [CDS (I) Paper 34 Q55]

(a) 40 cm (b) 48 cm (c) 50 cm (d) 60 cm

**570.** ABC is a triangle right-angled at B. Points M and N trisect BC. If AM = 6 cm and AN = 9 cm, then what is AC equal to ? [CDS (I) Paper 34 Q56]

(a) 4xfiö cm (b) 265 cm (c) 24 cm (d) 20 cm

**571.** The perimeter of the triangle is Select the answer using the code given below : [CDS (I) Paper 34 Q11]

(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11 23

**572.** Area of triangle OAC is equal to 102 cm2. Select the answer using the code given [CDS (I) Paper 34 Q11]

(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

## SECTION : GEOMETRY: CIRCLES, CHORDS & TANGENTS (29 QUESTIONS)

**573.** What is the of centres of circles which touch a given line at a given point ? [CDS (I) Paper 3 Q2]

(a) A line pen*ndicular to thc given line, passing through the given (b) A line parallel to the given line

(c) A circle tangent to the given line at the given point (d) A closed curve other -than a circle

**574.** If ABC is a triangle right angled at C and having u units, v units, w units as the lengths of its sides opposite to the veltices A, B, C respectively, then what is the tangent of the angle at A + tangent of the angle at B equal to ? (d) [CDS (I) Paper 3 Q37]

(a) 4 days (b) 5 days (c) 6 days (d) days 33. If of

**575.** 31-5 km/h 365 km/h 38-5 krn/h 40-5 km/h c In the figure given above, the side of square ABCD is 7 cm. What is the area of the shaded portion, formed by the arcs of the circles with centre at C and A ? [CDS (I) Paper 3 Q69]

(a) 7 crn2 (b) 28 cm2 (c) 14 cm2 (d) 21 Crn2 - A 20

**576.** A wire Of length 36 cm is bent in the form of a semicircle. What is the radius Of the semicircle? [CDS (I) Paper 4 Q74]

(a) 9 Crn (b) 8 Crn (c) 7 Crn (d) 6 Ctn

**577.** A circle and a souare have the same perimeter. Which one of the following is correct? [CDS (I) Paper 4 Q86]

(a) The area Of the circle is e qual to that of square

(b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square

(d) No conclusiion can be drawn 87* If (x 4 + x - 32? What is One of the values of (x (a) 18 (b) 16

**578.** The opposite angles of a cyclic quadrilateral are supplementary. Angle subtended by an arc at the centre is double the angle subtended by it at any point On the remaining part Of the circle. Which one of the following is correct in respect of the above statements ? [CDS (I) Paper 6 Q2]

(a) Statement- I statement-2

(b) Statement-2 statement-I

(c) Statement-I statement-2

(d) Neither statement-I statement-2 nor statement-2 statement-I (a) (c) (d)

**579.** The four points B, C, E and D may lie on a circle. Which of the above Statements is/are 95. (b) 17/8 correct ? [CDS (I) Paper 7 Q2]

(a) I only

(b) 2 only

(c) Both I and 2

(d) Neither 1 nor 2 and 81 equal to ? (a) (c) (d) 17/16 17/4 15/4 3, then what is x (a)

**580.** Area of triangle ABC 4 (Area Of tri angle BPQ). Select the correct answer using the code given [CDS (I) Paper 11 Q2]

(a) 1 only

(b) 2 only

(c) Both I and 2

(d) Neither I nor 2 AB and CD are two chords Of a circle meeting externally at P. Then Which of the following Ware correct ?

**581.** The chord of an arc of a circle is of length x, the height of the arc is y and the radius of the circle is z. Which one Of the following is correct? y(2z - y) xa y(2Z - y) = 4X2 (b) 2ygz-y1 x2 (c) 4yPz - x2 (d) . In a triangle ABC, ZB-2ZC=2ZA. 80 [CDS (I) Paper 12 Q79]

(a) 2 only above statements (b) AC/5 (c) AC/3 (d) AC/4

**582.** From an external point p tangents PA and PB are drawn to the circle as shown in the above figure. CD is the tangent to the circle at E. If AP 16 cm, then the perimeter of the triangle PCD is equal to [CDS (I) Paper 14 Q82]

(a) 24 cm (b) 28 cm (c) 30 cm (d) 32 cm 28

*— SPACE FOR ROUGH WORK —*

**583.** Chord CD intersects the diameter AB of a circle at riW1t angle at a m)int p in the ratio I : [CDS (I) Paper 14 Q84]
(a) 20 square t
(b) 30 square unit
(c) 40 square unit
(d) None of the above 84. Chord CD intersects the diameter AB of a circle at riW1t angle at a m

**584.** AD i s the diameter Of a rcle with area 707m2 and BC- as shown in the figure above. All curves inside the circle are semicirc s with their diameters on AD. What is the cost Of levelling the shaded regio at the rate of 63 per square metre? (d) e 29,700 e 22,400 14,847 None Of the above [CDS (I) Paper 15 Q100]
(a) AM < Median (b) GM Median
(c) AM = Me dian Suppose for Which one of the corre ct? (d) GM>1/6

**585.** A circle and a square have the same perimeter. Which one of the following is correct ? [CDS (I) Paper 16 Q54]
(a) Their areas are equal (b) The area Of the circle is larger
(c) The area of the square is - times area of circle (d) The area of the square is It times area of circle

**586.** ABCDA is a con-cyclic quadrilateral of a circle ABCD with radius r and centre at O If AB is the diameter and CD is parallel and half of AB and if the clrcle completes one rotation about the centre O, then the locus of the middle point of CD is a circle of radius 3r 2r - A 20 2-vfir 3 2 , then what is [CDS (I) Paper 16 Q70]
(a) 4275
(b) 4350
(c) 4375
(d) 4450 68. What is the area of a triangle with sides Of length 12 cm, 13 cm and 5 cm ?

**587.** A piu of wim of 33 is into an rc of a ofradius 14 is the angle by am at ca&e of circle ? (c) 1350 (d) 1500 C - SDHY T-MTÄ [CDS (I) Paper 21 Q54]
(a) 32%
(b) 30% Sl. If the Of h+ue of a trian* is, IO is tie of awh a riørt mgld ?
(c) cm2
(d) 50

**588.** Two circles of radii 20 cm and 16 cm intersect and the length of common chord is 24 cm. If d is the distance between their centres, then which one Of the following is correct ? 23 [CDS (I) Paper 22 Q69]
(a) d < 26 cm (b) 26 cm < d < 27 cm (c) 27 (d) d > 28 cm

**589.** In a circle of radius 5 cm, AB and AC are two chords such that AB 8 cm. What is the length of chord BC? [CDS (I) Paper 22 Q70]
(a) +1 (b) + 3 (c) l (d) 266+1

**590.** Two circles touch internally. The sum of their areas is 1367t cm2 and distance between their centres is 4 What are the [CDS (I) Paper 22 Q11]
(a) +1 (b) + 3 (c) l (d) 266+1

**591.** A circle of diameter 8 cm is placed in such a manner that it touches two perpendicular lines. Then another smaller circle is placed in the gap such that it touches the lines and the circle. What is the diameter of the smaller circle ? [CDS (I) Paper 22 Q73]
(a) - *E (b) litre (c) litre , (d) - litre 10

**592.** If an arc of a circle of radius 6 cm subtends a central angle measuring 300, then which one of the following is an approximate length of the arc ? [CDS (I) Paper 22 Q84]
(a) - AB (b) a AB2 (c) 1, 2,3 82. 83. 2 (d) 4

**593.** What is the area of the shaded region in the given figure, ifthe radius of each Of the circles is 2 cm ? [CDS (I) Paper 22 Q91]
(a) - cm2 (b) - cm2 (c) - E cm2 (d) cm2

**594.** What is the approximate area of minor segment of the circle ? [CDS (I) Paper 25 Q79]
(a) 2*71 cm2 (b) 2*42 cm2 (c) 1*91 cm2 (d) 1*71cm2 A -

**595.** What is the sum of the radii of the circles ? [CDS (I) Paper 25 Q81]
(a) 104 cm (b) 11-2 cm (c) 12 cm (d) 13 cm

**596.** If two chords AB and AC Of a circle are equal, then the centre Of the circle lies on the angle bisector Of angle CAB. If two concentric circles are intersected by a line at A, B, C and D respectively, then AC BD. Which Of the above statements is/are correct? [CDS (I) Paper 26 Q2]
(a) I Only (b) 2 only (c) Both I and 2 (d) Neither I nor 2

**597.** The area of the circle circumscribing three identical circles touching each other is square cm. What is the radius of one of the smaller circles ? [CDS (I) Paper 31 Q84]
(a) cm (b) I cm (c) cm (d) cm A -

*— SPACE FOR ROUGH WORK —*

**598.** A circle is inscribed in a triangle ABC right-angled at B. If AB 5 cm and BC = 12 cm, then what is the radius of the circle ? [CDS (I) Paper 31 Q94]

(a) 121 (b) 120

(c) 119 (d) None of the above 91. What is the HCF 0f236-1 and 245-1 ?

**599.** In a circle of radius 14 cm, APB is a shorter arc and P is the midpoint of the arc. Ikt C be the midpoint Of the chord AB and PC 7 cm. What is the length Of chord AP ? the (d) 14 cm 1 P.T.O. [CDS (I) Paper 33 Q65]

(a) 20%

(b) 40%

(c) None of the above 61. Let AD be the altitude of a triangle ABC If and

(d) I only

**600.** The diagonal BD is parallel to the tangent of the circle at A Select the answer using the code given below : [CDS (I) Paper 34 Q11]

(a) I only (b) 11 only (c) Both I 11 (d) Neither I nor 11

**601.** Let O be the centre of a circle. Let A, B and C lie on the circle such that LABC= 1530. What is the value of LAOC ? 21 [CDS (I) Paper 34 Q52]

(a) 540 (b) 520 (c) 390 (d) 270

## SECTION : MENSURATION 2D: AREAS & PERIMETERS (47 QUESTIONS)

**602.** ABCD is a square. p. Q, R, S are points on the sidesAB, BC, CD, DA respectively such that CR=DS. What is ZSPQ equal ? [CDS (I) Paper 3 Q49]

(a) 300 (b) 450 (c) 600 (d) 900

**603.** A parallelogram and a rectangle stand on the same base and on the same side of the base with the same height. If 11, be the perimeters of the parallelogram and the rectangle respectively, then which one of the following is correct ? [CDS (I) Paper 3 Q58]

(a) 11 12 (b) 11 = 12 (c) 12 but 11 *212 (d) 11 -212

**604.** Which one of the following numbers is not a square of any natural ? [CDS (I) Paper 3 Q88]

(a) 5041 (b) 9852 (c) 1936 (d) 6241 T-DETA.K.FN - A 26

**605.** What is the maximum area of a rectangle, the IVimeter of which is 18cm? [CDS (I) Paper 3 Q89]

(a) 2025 cm2 (b) 20-00 cm2 (c) 19-75 cm2 (d) 19-60 cm2

**606.** The area of a rectangle whose length is 5 units more than twice its width is 75 square units. What is its width ? [CDS (I) Paper 6 Q71]

(a) 3 units (b) 5 units (c) 7 units (d) IO units

**607.** The diameter Of a circle circumscribing a Square is 1545 cm. What is the length of the side of the square ? [CDS (I) Paper 6 Q100]

(a) 360 450

(b) 48

(c) 600 97. 470 670 N In the figure given above, PQ is parallel to RS. What is Z NMS equal to ?

(d) 270

**608.** The area of a rectangle whose length is 5 more than twice its width is 75 square units. What is the perimeter of the rectangle ? [CDS (I) Paper 7 Q59]

(a) 40 units (b) 30 units

(c) 24 units (d) 20 units Two number* in the ratio 2

**609.** The Of y if one Of the Of the - bx + c O the square at the then Which one the following is ? [CDS (I) Paper 9 Q21]

(a) 120 $cm^3$ (b) 150 $cm^3$ (c) 180 $cm^3$ (d) 216 $cm^3$

**610.** The perimeter of a rectangle having area equal to 144 cm and sides in the ratio 4 : 9 is 52 cm (b) 56 (c) 60 cm (d) 64 cm [CDS (I) Paper 10 Q46]

(a) 56 (b) 60 cm (c) 64 cm (d) 1 only

**611.** What is the area between a square Of side IO cm and two inverted semicircular cross-sections each of radius 5 cm inscribed in the square? [CDS (I) Paper 10 Q48]

(a) 56 (b) 60 cm (c) 64 cm (d) 1 only

**612.** Four equal-sized maximum circular plates are cut off from a square paper sheet Of area 784 square Cm. The circumference of each plate is [CDS (I) Paper 13 Q57]

(a) I Ctn (b) 22 cm

(c) 33 cm (d) 44 W/ 52A 20 58. is equal to - I, then (a) 0

**613.** From a circular piece of cardboard of radius 3 cm, two sectors of4(r each have been cut Off. The area Of the remaining portion is [CDS (I) Paper 13 Q89]

(a) I I square cm (b) 22 square cm (c) 33 square cm (d) 44 square cm

*— SPACE FOR ROUGH WORK —*

**614.** A square and an equilateral triangle have equal perimeter. If the diagonal of .the square is 12-45 cm, then the area of the triangle is [CDS (I) Paper 14 Q54]

(a) cm2 (b) 240 CITI 2 (c) 480 cm2 (d) 6443 cm2 18

**615.** If X is any point within a square ABCD and on AX a square AXYZ is described, which Of the following is/are corect ? 2 ZABX=ZADZor ZADX= ZABZ Select the correct answer using the code given txlow [CDS (I) Paper 14 Q67]

(a) I only (b) 2 only (c) Both 1 and 2 (d) Neither I nor 2 - A 24

**616.** Three rectangles RI, Ra and Rs have the same area. Their lengths x, , xz 'and x, respectively are such that x, < x, < xy If VI , V2 and V3 are the volumes of the cylinders formed from the rectangles RI, R2 and Rs respectively by joining the parallel sides along the breadth, then which one of the following is correct ? [CDS (I) Paper 14 Q76]

(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

**617.** DP and when produced meet at Q. If area of triangle BPQ is 20 square unit, the area of triangle DC? is [CDS (I) Paper 14 Q2]

(a) 20 square t (b) 30 square unit (c) 40 square unit (d) None of the above

**618.** A rhombus is formed by joining midpoints of thc sides of a rectangle in the suitable order. If the area of the rhombus is 2 square units, then the area Of the rectangle is [CDS (I) Paper 15 Q53]

(a) 242 square units (b) 4 square units (c) 40 square units (d) 8 square units

**619.** The base Of an isosceles triangle is 300 units and each of its equal sides is 170 units. Then the area Of the triangle is [CDS (I) Paper 15 Q57]

(a) 9600 square units (b) 10000 square units (c) 12000 square units (d) None of the above

**620.** Four equal discs are placed such that each one touches two others. If the area Of empty space enclosed by them is 150/847 square centimetre, then the radius of each disc is equal to [CDS (I) Paper 15 Q58]

(a) 7/6 cm (b) 5/6 cm (c) 112 cm (d) 5/11 cm

**621.** The cost of painting a spherical vessel of diameter 14 cm is Rs. 8008. What is the cost Of painting per square centimetre ? [CDS (I) Paper 16 Q52]

(a) Rs. 8 (b) Rs. 9 (c) Rs. 13 (d) Rs. 14

**622.** What is area of. largest triangle inscribed in a semi-circle Of radius r units ? [CDS (I) Paper 16 Q62]

(a) square units (b) 2r2 square units (c) 3? square units (d) 4r2 square units

**623.** How many square metres Of canvas are used ? [CDS (I) Paper 16 Q66]

(a) 14450 (b) 14480 (c) 14580 (d) 14850 18

**624.** Unit in is 7. of the square of two is always divisi- by [CDS (I) Paper 21 Q3]

(a) be 2 one Of the 61%wing ? 13. HCF numbes is 12. Wych me Of xver LCM ?

(b) 80

(c) 60 36 *

(d) 24 14. WEch Of

**625.** Unit in is 7. of the square of two is always divisi- by [CDS (I) Paper 21 Q2]

(a) be 2 one Of the 61%wing ? 13. HCF numbes is 12. Wych me Of xver LCM ?

(b) 80

(c) 60 36 *

(d) 24 14. WEch Of

**626.** If the area of a square is 2401++ 196r2+4, then what is its side length ? [CDS (I) Paper 22 Q23]

(a) 49x2 -f- 3x + 2 (b) 49x2 _ 3x + 2 (c) 49++2 (d) 59++2

**627.** In a pi-diagram there are three sectors. If the ratio Of the angles Of the sectors is 1 : 2 : 3, then what is the angle of the largest sector ? [CDS (I) Paper 22 Q36]

(a) 2000 (b) 1800 (c) 1500 (d) 1200

**628.** If area of a circle and a square are same, then What is the ratio Of their perimeters ? 2 4 A - [CDS (I) Paper 22 Q72]

(a) +1 (b) + 3 (c) I (d) 266+1

**629.** A square and a rectangle have equal areas. If one side of the rectangle is of length numerically equal to the square of the length of the side of the square then the other side of the rectangle is 25 25 [CDS (I) Paper 22 Q75]

(a) square root of the side of the (b) half the side of the square

(c) of unit length (d) double the side of the square

**630.** Suppose a region is formed by removing a sector of 200 from a circular region of radius 30 feet. What is the area of this new region ? [CDS (I) Paper 22 Q77]

(a) 1507t square feet (b) 550z square feet (c) square feet (d) 850% square feet A -

**631.** Area of APDA is equal to that of APCQ [CDS (I) Paper 22 Q1]

(a) - AB (b) a AB2 (c) 1, 2,3 82. 83. 2 (d) 4

*— SPACE FOR ROUGH WORK —*

**632.** Area of AQAB is equal to twice that of APCQ Select the correct answer using the code given below : [CDS (I) Paper 22 Q2]
(a) 1 only (b) 2 only (c) Both I and 2 (d) Neither 1 nor 2

**633.** What is the area of the largest square plate cut from a circular disk Of radius one unit ? [CDS (I) Paper 22 Q86]
(a) 4 square units (b) 245 square units (c) r square units (d) 2 square units

**634.** cm What is the approximate area of the shaded region in the figure given ? 35 [CDS (I) Paper 22 Q99]
(a) 15*3 cm2 (b) 25*5 cm2 (c) 284 cm2 (d) cm2

**635.** A persm wishes to fence 375 m2 rectangular garden. He has 65 m of barbed wire and is able to fence only three sides of the garden. What is the perimeter of the garden ? [CDS (I) Paper 25 Q30]
(a) 80 m (b) 84 m (c) 90 m (d) 100m

**636.** Let p be the area of a square X and q be the area of the square formed on the diagonal of the square X. What is the value of - ? [CDS (I) Paper 25 Q63]
(a) sin 9
(b) cose
(c) tan 0
(d) cot O 59. If A, B and C are interior angles of a triangle ABC, then what is equal to ? 2 A+B+C

**637.** The area of a rhombus is 336 square cm. Ifthe length ofone of its diagonals is 48 cm, then what is the perimeter of rhombus ? [CDS (I) Paper 25 Q64]
(a) 200 cm (b) 120 cm (c) IOOcm (d) 90 cm A - 2 sin39 - sino

**638.** What is the least value of n if 194480+nzm4, where m and n are natural numbers? If then What is the value Of ? What is the area Of the region enclosed by three identical circles (each of radius 4 cm) touching each other? [CDS (I) Paper 26 Q18]
(a) - square cm (b) 16vTä - - square cm (c) (160 - 8r) square cm (d) - square cm

**639.** ABCD is paralle I square metre square metre square metre square metre a trapezium in which AB is to DC Let E and F be the midpoints on AD and BC respectively. If EF IO cm and AB - DC 4 cm, then what is the value of ABX DC? [CDS (I) Paper 26 Q73]
(a) 84 square cm (b) 96 square cm (c) 100 square cm (d) 108 square cm

**640.** If the perimeter of an isosceles right triangle is 4 (2 cm, then What is its area in square cm? The 8 16 24 diagonal of a square is 120 cm and the area of an equilateral triangle is 64" Square Cm. Which Of the following * statements is/ are correct? part? [CDS (I) Paper 26 Q77]
(a) 150 (b) 225 (c) 300 (d) 450

**641.** A square is drawn inside a square of side 14 cm in such a way that the corners of the inner square coincide with the mid points of the sides of the outer square. What is the area lying between the two squares ? [CDS (I) Paper 31 Q48]
(a) 98 square cm (b) 56 square cm (c) 49 square cm (d) 24*5 square cm

**642.** What is the area of triangle OMN? [CDS (I) Paper 31 Q60]
(a) 36 square cm (b) 40 square cm 21 21 (c) 45 square cm (d) 48 square cm

**643.** What is the maximum area of a rectangle, in square cm, whose perimeter is 400 cm ? [CDS (I) Paper 31 Q72]
(a) 100 (b) 200 (c) 1000 (d) 10,000

**644.** The area included between nth square and (n - l)th square is independent of n 35 Which of the statements given above is/ are correct ? [CDS (I) Paper 31 Q11]
(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

**645.** In a rectangle ABCD, AC is one of the diagonals. If AC+AB=3AD and AC - AD 4 units, then what is the area of the triangle ? [CDS (I) Paper 31 Q83]
(a) 24 square unit (b) 36 square unit (c) 48 square unit (d) 72 square unit

**646.** In a triangle ABC, 21 cm, BC - 20cm and CA = 13 cm. A per- pendicular CD is drawn upon the longest side. What is the area of the triangle BCD ? [CDS (I) Paper 31 Q85]
(a) 96 square cm (b) 84 square cm (c) 80 square cm (d) 72 square cm

**647.** Q can be a perfect square. Q is positive for all real values of x. 9 Select the answer using the code given below : [CDS (I) Paper 34 Q11]
(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

**648.** A metallic sheet of rectangular shape is 48 cm x 36 cm. From each corner a square of side x cm is cut off and an open box is made of the remaining sheet. If the box holds 5*12 litre of water, then what is the value of x ? [CDS (I) Paper 34 Q66]
(a) 6 cm (b) 7-5 cm (c) 8 cm (d) 8-5 cm 27

## SECTION : MENSURATION 3D: SURFACE AREAS & VOLUMES (35 QUESTIONS)

**649.** Three cubes each Of side 5 cm are joined end to end. What is the surface area Of the resulting cuboid ? [CDS (I) Paper 1 Q50]
(a) 300 cml (b) 350 cm2 (c) 375 cm2 (d) 400 cm2

*— SPACE FOR ROUGH WORK —*

**650.** What is volume of the frustum of a cone with height h and radii r, , ? [CDS (I) Paper 3 Q74]
(a) 4-5 74. What is volume of the frustum of a cone with height h and radii r, , ?
(b) 9th
(c) 15m
(d) I *5m

**651.** A cylinder having base of circumference 60 cm is rolling without sliding at a rate of 5 rounds per second. How much dis- tance will the cylinder roll in 5 seconds ? [CDS (I) Paper 3 Q75]
(a) 4-5 74. What is volume of the frustum of a cone with height h and radii r, , ?
(b) 9th
(c) 15m
(d) I *5m

**652.** A hemisphere is made of a sheet of a metal 1 cm thick. If the outer radius is 5 what is the weight of the hemi- sphere(l cm3 of the metal weighs 9 g) ? [CDS (I) Paper 3 Q79]
(a) 54 g (b) 366 g (c) 122 g (d) 1087t g

**653.** If three cubic biscuits having edges m, m and 0-5 m respectively are melted and formed into a single cubic biscuit, then What is the total surface area Of the cubic biscuit? [CDS (I) Paper 4 Q72]
(a) rn2 (b) 1-56 rn2 (c) 1-84 m2 (d) 2*16 m 2

**654.** The material Of a solid cone is converted into the shape of a solid cylinder Of equal radius. If the height of the cylinder is 5 cm. what is the height Of the cone ? [CDS (I) Paper 5 Q91]
(a) 15 cm (b) 20 cm (c) 25 cm (d) 30 cm

**655.** A Wire of diameter 5 mm is wound around it, so as to cover it completely. What is the length of the wire used ? [CDS (I) Paper 5 Q94]
(a) 22 m (b) 20 m (c) 18 m (d) None Of the above

**656.** If the diameter of a sphere is doubled, how does its surface*area change ? [CDS (I) Paper 6 Q72]
(a) It increases two times. (b) It increases three times. (c) It increases four times, (d) It increases eight times.

**657.** What is the number of wax balls, each Of radius I cm, that can be molded out of a sphere of radius 8 cm ? [CDS (I) Paper 6 Q80]
(a) 256 (b) 512 (c) 768 (d) 1024

**658.** A cardboard sheet in the form of a circular sector of radius 30 cm and central angle 1440 is folded to make a cone. What is the radius of the cone ? [CDS (I) Paper 7 Q46]
(a) 12 cm (b) 18 cm (c) 21 cm (d) None Of the above

**659.** 20 What is the difference in radii of the ends of the frustum ? (b) (d) None of the above What is the volume ofthe frustum ? [CDS (I) Paper 7 Q73]
(a) 12
(b) 13 If 196x4 = x6, then x3 is equal to which one of the following ?
(c) x'/ 14
(d) 14xa

**660.** Let be a pyramid on a square base and B be a cube. Let a, b, c denote the number of edges, number of faces and number of corners respectively. Then the result a = b + c is true for [CDS (I) Paper 10 Q47]
(a) A only (b) B only (c) both A and B (d) neither A nor B

**661.** What is the surface area of the double cone so formed? (c) 1310-4 cm 2 (d) 13188 cm2 [CDS (I) Paper 12 Q88]
(a) The triangle is isosceles (b) The triangle is equilateral
(c) None Of the ab ove For the next three (d) 3124

**662.** If three metallic spheres Of radii 6 8 cm and IO cm are melted to form a single sphere, then the diameter Of the new sphere will be [CDS (I) Paper 13 Q46]
(a) 12 Cm (b) 24 cm (c) 30 cm (d) 36 Crn

**663.** A large water tank has the shape of a cube. If 128 mJ of water is pumped out, the water level goes down by 2 m. "men the maximum capacity of the tank is [CDS (I) Paper 14 Q60]
(a) 512 m3 (b) 480 ms (c) 324 (d) 256 my

**664.** The areas of three mutually perpendi- cular faces of a cuboid are x, y, z. If V is the volume, then xyz is equal to [CDS (I) Paper 14 Q74]
(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

**665.** Let V be the volume of an inverted cone with vertex at origin and the axis of the cone is along positive y-axis. 'Ihe cone is filled with water up to half of its height. The volume of water is 8 6 2 [CDS (I) Paper 14 Q75]
(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

*— SPACE FOR ROUGH WORK —*

**666.** Ifthesurface area ofa cube is 13254 cm2, then the length of its diagonal is [CDS (I) Paper 14 Q77]
(a) 4445 cm (b) 44-16 cm (c) 4745 cm (d) 476 cm

**667.** If the total surface area is 616 cm2, what is the volume of the cylinder ? [CDS (I) Paper 16 Q2]
(a) 539 cm3 (b) 616 cm3 (c) 1078 cm3 (d) 1232 cm3

**668.** A drinking glass Of height 24 cm is in the shape of frustum of a cone and diameters of its bottom and top circular ends are 4 cm and 18 cm respectively. If we take capacity Of the glass as xx cm , then what is the value Of x ? [CDS (I) Paper 16 Q55]
(a) 824 (b) 1236 (c) 1628 (d) 2472

**669.** The height of a cone is 60 cm. A small cone is cut Off at the top by a plane parallel to the base and its volume is -L the volume of original cone. What is the height from the base at which the section is made ? (b) 20 cm (c) 30 cm (d) 45 cm [CDS (I) Paper 16 Q58]
(a) Rs. 8
(b) Rs. 9
(c) Rs. 13
(d) Rs. 14 53. If a quadrilateral has an Inscribed then the sum of a pair Of opposite sides equals

**670.** What is the slant height of the cone ? [CDS (I) Paper 16 Q64]
(a) 60 m (b) 65 m (c) 68 m (d) 70m

**671.** A cinular a Of 20 cm its curved na is 1000 cml. What is ere volume Of [CDS (I) Paper 21 Q53]
(a) 40m cm' (b) 4500 cm3 (c) 5000 cm3 (d) cm3

**672.** The thickness of a cylinder is I foot, the inner radius of the cylinder is 3 feet and height is 7 feet. TO paint the inner surface it requires one litre of a particular colour. How much quantity of the same colour is required to paint all the surfaces of the cylinder ? [CDS (I) Paper 22 Q74]
(a) litre (b) litre , (c) - litre 10 (d) - litre

**673.** A right circular cylinder just encloses a sphere. If p is the surface area of the sphere and q is the curved surface area of the cylinder, then which one of the following is correct ? (b) p 24 (c) 2p = q [CDS (I) Paper 22 Q88]
(a) 7-5 m 86. What is the area of the largest square plate cut from a circular disk Of radius one unit ?
(b) 4 square units
(c) 245 square units
(d) r square units

**674.** If the volume of the cylinder is 1617 CITI 3 , then what is the curved surface area of the cylinder? (Take It = - I [CDS (I) Paper 24 Q3]
(a) 242 cm (b) 385 cm 2 (c) 462 cm 2 (d) 770 cm

**675.** A cone of height 16 cm and diameter 14 cm is mounted on a hemisphere of same diameter. What is the volume of the solid thus formed ? (take [CDS (I) Paper 25 Q69]
(a) 1540 cm3 (b) 1078cm3 (c) 1048 cm3 (d) 770 cm3

**676.** cubes each of volume 343 cm3 are joined end to end. What is the total surface area of the resulting cuboid ? [CDS (I) Paper 25 Q70]
(a) 343 cm2 (b) 350 cm2 (c) 686 cm2 (d) 700 cm2

**677.** A cubical block of side 14cm is sur- mounted by a hemisphere of radius 7 cm. What is the total surface area of the solid thus formed ? (take 9) [CDS (I) Paper 25 Q71]
(a) 1330 cm2 (b) 1306 cm2 (c) 1296 cm2 (d) 1256 cm2

**678.** How many silver coins, cm in diameter and of thickness 4 mm, must be melted to form a cuboid of dimen- sions 21 cm x 11 cm x 7 cm ? (take 4) [CDS (I) Paper 25 Q72]
(a) 420 (b) 210 (c) 200 (d) 168 A -

**679.** A tub is in the sharx of a frustum Of a cone. The radii of two circular ends of the tub are 105 cm and 42 cm. If the vertical height of the tub is 16 cm, what is its slant height ? [CDS (I) Paper 25 Q73]
(a) 63*5 cm (b) 65 cm (c) 73-5 cm (d) 75 cm

**680.** The sum of the height and the radius Of a right circular cylinder is 2 1 cm, and the radius is less than the height. If the curved surface area of the cylinder is 616 cm2, then What is the volume Of the cylinder? (Take It 22 / [CDS (I) Paper 33 Q62]
(a) 1078 cm (b) 1617 cm (c) 1927 cm 3 (d) 2156 cma

**681.** The length, breadth and height Of a cuboid are consecutive integers. If the volume of the cuboid is 336 cubic units, then what is the total surface area Of the cuboid? [CDS (I) Paper 33 Q64]
(a) 288 square units (b) 292 square units
(c) 296 square units (d) Cannot be determined due to insuffcient data

**682.** A sphere of diameter 14 cm is dropped into a cylindrical vessel partly filled with water. The radius of the vessel is 14 cm. If the sphere is completely submerged in water, then how much will the level of water rise ? 7 [CDS (I) Paper 34 Q58]
(a) - cm (b) - cm 14 (c) - cm 16 (d) - cm

*— SPACE FOR ROUGH WORK —*

**683.** Let x, y, z (all prime) be the length, breadth and height respectively of a cuboid with x > y > z. The volume of the cuboid is 30k3 cubic units, where k is a natural number. What is the total surface area of the cuboid ? [CDS (I) Paper 34 Q70]

(a) 60 square units (b) 62 square units

(c) 64 square units (d) Cannot be determined due to insufficient data

## SECTION : STATISTICS & DATA INTERPRETATION (24 QUESTIONS)

**684.** If each number is first multiplied by 2 and then increased by 5, then what is the mean Of new numbers ? [CDS (I) Paper 1 Q20]

(a) 20 (b) 25

(c) 40 (d) 45 Consider the following types of data

**685.** Marks of students who appeared for a test of 100 marks. Collar sizes of200 shirts sold in a week. Monthly incomes of250 employees Of a factory. For which of the above data, mode is a suitable measure of central tendency ? [CDS (I) Paper 1 Q2]

(a) I and 2 only (b) 2 Only (c) I and 3 only (d) 1 , 2 and 3

**686.** What is the mean of the ages of minor children ? [CDS (I) Paper 3 Q7]

(a) 3 years (b) 4 years (c) 5 years (d) 6 years

**687.** Frequency polygon Frequency curve Frequency distribution All of the above D E Three straight lines X, Y and Z are parallel and the angles are as shown in the figure above. What is ZAFB equal to ? [CDS (I) Paper 3 Q83]

(a) 200 (b) 150

(c) 300 (d) 100 - A 24 (a) (b) (c) (d) (a) 1 (b) 2 (c) 3 (d) 4

**688.** Consider *the following frequency distributién * Class 0-10 0-20 0-30 0-40 0-50 Frequency 3 What is the above frequency distributioÖ known as? [CDS (I) Paper 4 Q82]

(a) Cumulative distribution in more than type (b) Cumulåtive distribution in less than type

(c) Continuous frequency distri- bu tion (d) None Of above / 26A 26

**689.** Histogram is an equivalent graphical representation of the frequency distribution. Histogram is suitable for continuous random variables, where the total frequency Of an interval is evenly distributed over the interval. Which . of the statements given above is/ are correct? [CDS (I) Paper 4 Q2]

(a) only

(b) 2 only

(c) Both I and 2

(d) Neither I nor 2 What is the median Of the values * 11, 7, 6, 9, 12, 15, (b) 11 (c) 12 (d) 15

**690.** X is as given below : x frequency [CDS (I) Paper 6 Q67]

(a) 36 (b) 44 (c) 52 64. (d) 150

**691.** The frequency distribution of a variate X is as given below : [CDS (I) Paper 6 Q69]

(a) 36 (b) 44 (c) 52 64. (d) 150

**692.** If 15 is added to each of the first forty values and 5 is subtracted from each of the rernaining sixty values, the new mean becomes : [CDS (I) Paper 8 Q45]

(a) 45 (b) 48 (c) 51 (d) 55 28

**693.** Consider the following distribution Value of the variable 1 2 3 4 5 naaaa Frequency For What value Of f, is the arithmetic mean of the above distribution 3.1 ? [CDS (I) Paper 8 Q100]

(a) Mean Median (b) Mean Mode

(c) Mean > Median (d) Mean > Mode What is the harmonic mean 25, 40 and 50 ?

**694.** If each of n numbers x, - i (i = l, 2, 3, ...m) is replaced by (i + l)xi, then the new mean IS n(n + l) 2 (d) [CDS (I) Paper 14 Q99]

(a) 24 (b) a - 25 (c) 28 (d) a - 1 25

**695.** The weighted arithmetic mean of first 10 natural numbers whose weights are equal to the corresponding numbers is equal to (d) 385 7 14 35 [CDS (I) Paper 14 Q100]

(a) 24 (b) a - 25 (c) 28 (d) a - 1 25

**696.** The election result in Which six parties contested was depicted by a pie chart. Party A had an angle 135E on this pie chart. If it secured 21960 votes, how many valid votes in total were cast? [CDS (I) Paper 15 Q91]

(a) 51240 (b) 58560 (c) 78320 (d) 87840

**697.** If the mean age of combined group Of boys and girls is 18 years and-the mean Of age of boys is 20 and that of girls is 16; then what is the percentage of boys in the group ? [CDS (I) Paper 16 Q40]

(a) 60 (b) 50 (c) 45 (d) 40 - A 12

**698.** If x is the mean Of the numbers in the first pair and y is the mean of the numbers in the second pair, then what is the mean of x and y ? [CDS (I) Paper 22 Q12]

(a) 60 (b) 72 (c) 75 (d) 78

*— SPACE FOR ROUGH WORK —*

**699.** For two observations, the sum is S and product is P. What is the harmonic mean Of these two observations ? (b) S A - [CDS (I) Paper 22 Q45]

(a) A, B and C
(b) A, B and D
(c) A, E and F 41. The number of three digit numbers
(d) Six

**700.** What is the arithmetic mean of the first ten composite numbers? 102 (d) 112 [CDS (I) Paper 24 Q94]

(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015?
(b) and E
(c) A, C and D only
(d) C D and E

**701.** What is the mean of the values ? [CDS (I) Paper 27 Q70]

(a) 38 (b) 40 (c) 42 (d) 44 GI. 62. 63. 64.

**702.** The arithmetic mean of n numbers is M. If the sum of first (n - 1) terms is k, then what is the nth number ? (b) nM - k (c) n(M - k) [CDS (I) Paper 31 Q21]

(a) nM - k (b) n (c) 81 (d) 105

**703.** What is the geometric mean of 3, 9, 27, 81, 243, 729, 2187? [CDS (I) Paper 31 Q22]

(a) 81 (b) 105 (c) 144 (d) 243

**704.** If 5 is multiplied to each observation, then what Will be the new arithmetic mean? [CDS (I) Paper 33 Q60]

(a) 500 (b) 300 (c) 60 (d) 40

**705.** A distribution consists of 3 components With frequencies 45, 40 and 55 having their means 2, 2 * 5 and 2 respectively. What is the mean of the combined dis tribu tion? [CDS (I) Paper 33 Q95]

(a) 2 14 (b) 225 (c) 2 37 (d) 2 50

**706.** If 10 is added to each observation then what will be the new arithmetic mean ? [CDS (I) Paper 34 Q50]

(a) 50 (b) 55 (c) 60 (d) 65 15 A -

**707.** What is the arithmetic mean ? [CDS (I) Paper 34 Q54]

(a) 3-25 (b) 3-50 (c) 3-75 (d) 4-25

*— SPACE FOR ROUGH WORK —*